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Long-term antibiotic prophylaxis for prevention of rheumatic fever recurrence and progression to rheumatic heart disease (Review)

Bray JJH, Thompson S, Seitler S, Ali SA, Yiu J, Salehi M, Ahmad M, Pelone F, Gashau H, Shokraneh F, Ahmed N, Cassandra M, Marijon E, Celermajer DS, Providencia R

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[Rapid Review]

Long-term antibiotic prophylaxis for prevention of rheumatic fever recurrence and progression to rheumatic heart disease

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ABSTRACT

Background

Rheumatic fever is a non-suppurative, inflammatory sequela of group A *Streptococcus* pharyngitis that can occur at two to four weeks after infection. Following an episode of rheumatic fever, there is a risk of developing rheumatic heart disease (RHD) later in life that carries significant risk of morbidity and mortality. RHD remains the largest global cause of cardiovascular disease in the young (age < 25 years). The historical literature provides inconclusive evidence that antibiotic prophylaxis is beneficial in reducing the risk of recurrence of rheumatic fever and development of RHD. Antibiotics are thought to work by reducing the carriage of group A *Streptococcus* and thus reducing the risk of infection. This review was commissioned by the World Health Organization (WHO) for an upcoming guideline.

Objectives

1. To assess the effects of long-term antibiotics versus no antibiotics (control) for secondary prevention of rheumatic fever recurrence and associated sequelae in people with previous rheumatic fever or RHD.

2. To assess the effects of long-term intramuscular penicillin versus long-term oral antibiotics for secondary prevention of rheumatic fever recurrence and associated sequelae in people with previous rheumatic fever or RHD.

Search methods

We systematically searched CENTRAL, MEDLINE, Embase, Conference Proceedings Citation Index-Science, clinical trial registers, ISRCTN.com and reference lists without restrictions on language or date up to 10 March 2024.

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Selection criteria

We sought randomised controlled trials or quasi-randomised trials, described in any language, including participants with previous rheumatic fever and/or RHD of any age, based in community or hospital settings. Studies were included if they compared firstly antibiotic prophylaxis with no antibiotic prophylaxis, and, secondly, intramuscular penicillin prophylaxis versus oral antibiotic prophylaxis.

Data collection and analysis

We used standardised methodological, Cochrane-endorsed procedures and performed meta-analyses with risk ratios (RR) and Peto odds ratios (Peto OR). Our primary outcomes were recurrence of rheumatic fever, progression or severity of RHD and cardiac complications. Our secondary outcomes were obstetric complications (maternal and foetal events), mortality, treatment adherence, adverse events and acceptability to participants. We performed comprehensive assessments of risk of bias and certainty of evidence, applying the GRADE methodology.

Main results

We included 11 studies (seven RCTs and four quasi-randomised trials) including 3951 participants. The majority of the included studies were conducted in the USA, UK and Canada during the 1950s to 1960s. Most participants with previous rheumatic fever had been diagnosed using the modified Jones criteria (mJC) (four studies), were an average of 12.3 years of age and 50.6% male. We assessed the majority of the included studies to be at high risk of bias, predominantly relating to blinding and attrition bias.

Comparison one: antibiotics versus no antibiotics

Pooled meta-analysis of six RCTs provides moderate-certainty evidence that antibiotics overall (oral or intramuscular) probably reduce the risk of recurrence of rheumatic fever substantially (0.7% versus 1.7%, respectively) (risk ratio (RR) 0.39, 95% confidence interval (CI) 0.22 to 0.69; 1721 participants). People with early or mild RHD likely have the greatest capacity to benefit from intramuscular antibiotic prophylaxis (8.1%) compared to no antibiotics (0.7%) (RR 0.09, 95% CI 0.03 to 0.29; 1 study, 818 participants; moderate-certainty evidence). Antibiotics may not affect mortality in people with late-stage RHD (RR 1.23, 95% CI 0.78 to 1.94; 1 study, 994 participants; low-certainty evidence). Antibiotics may not affect the risk of anaphylaxis (Peto odds ratio (OR) 7.39, 95% CI 0.15 to 372; 1 study, 818 participants; low-certainty evidence) or sciatic nerve injury (Peto OR 7.39, 95% CI 0.15 to 372; 1 study, 818 participants; low-certainty evidence) compared with no antibiotics, but probably have an increased risk of hypersensitivity reactions (RR 137, 8.51 to 2210; 2 studies, 894 participants; moderate-certainty evidence) and local reactions (RR 29, 1.74 to 485; 1 study, 818 participants; moderate-certainty evidence).

Comparison two: intramuscular antibiotics versus oral antibiotics

Pooled analysis of two RCTs showed that prophylactic intramuscular benzathine benzylpenicillin likely reduces recurrence of rheumatic fever substantially when compared to oral antibiotics (0.1% versus 1%, respectively) (RR 0.07, 95% CI 0.02 to 0.26; 395 participants; moderate-certainty evidence). Furthermore, it is unclear whether intramuscular benzyl penicillin is superior to oral antibiotics in reducing the risk of mortality in the context of RHD (Peto OR 0.22, 95% CI 0.01 to 4.12; 1 study, 431 participants; very low-certainty evidence). There were no data available on progression of latent RHD or adverse events including anaphylaxis, sciatic nerve injury, delayed hypersensitivity/allergic reactions and local reactions to injection.

Authors' conclusions

This review provides evidence that antibiotic prophylaxis likely reduces the risk of recurrence of rheumatic fever compared to no antibiotics, and that intramuscular benzathine benzylpenicillin is probably superior to oral antibiotics (approximately 10 times better). Moreover, intramuscular benzathine benzylpenicillin likely reduces the risk of progression of latent RHD. Evidence is scarce, but antibiotics compared with no antibiotics may not affect the risk of anaphylaxis or sciatic nerve injury, but probably carry an increased risk of hypersensitivity reactions and local reactions. Antibiotics may not affect all-cause mortality in late-stage RHD compared to no antibiotics. There is no evidence available to comment on the effect of intramuscular penicillin over oral antibiotics for progression of latent RHD and adverse events, and little evidence for all-cause mortality. It is important to interpret these findings in the context of major limitations, including the following: the vast majority of the included studies were conducted more than 50 years ago, many before contemporary echocardiographic studies; methodology was often at high risk of bias; outdated treatments were used; only one study was in latent RHD; and there are concerns regarding generalisability to low socioeconomic regions. This underlines the need for ongoing research to understand who benefits most from prophylaxis.

PLAIN LANGUAGE SUMMARY

Do long-term antibiotics help to reduce rheumatic fever recurrence and progression of rheumatic heart disease?

Key messages

In people who have had previous rheumatic fever (the body attacking itself in response to bacterial infection) or have rheumatic heart disease (long-term damage to the heart due to rheumatic fever):

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- long-term antibiotics (either injected into the muscle every month or taken as a tablet every day) probably reduce the risk of getting more episodes of rheumatic fever compared to no antibiotics;

- intramuscular antibiotics probably reduce the progression (getting worse) of early heart disease compared to no antibiotic; however, there is no evidence to compare intramuscular with oral antibiotics for progression of late-stage heart disease;

- antibiotics may not increase the risk of complications, such as a severe allergic reaction (anaphylaxis).

What is rheumatic heart disease?

Rheumatic heart disease is the top cause of heart disease in young people worldwide and kills about one-third of a million people every year. Rheumatic fever happens when the body's own defences go wrong, often because of a throat infection, fighting the heart instead of the bacteria. This can then lead to damage to the heart valves (gateways between rooms in the heart) called rheumatic heart disease. Antibiotics kill bacteria that can cause infections, reducing the risk of people developing rheumatic fever.

What did we want to find out?

We wanted to find out whether and, if so, how effective antibiotics are at reducing the chances of getting rheumatic fever again, and this leading to rheumatic heart disease.

What did we do?

We included studies that randomly gave people with past rheumatic fever or rheumatic heart disease antibiotics or not (e.g. based on flipping a coin). We were interested in comparing, firstly, long-term antibiotics with no antibiotics and, secondly, long-term intramuscular penicillin with long-term oral antibiotics. Participants in the studies we included had previous rheumatic fever or rheumatic heart disease, but could be any age. We looked for lots of different events that could have happened, including the rheumatic fever coming back (rheumatic fever recurrence), rheumatic heart disease getting worse (progression of rheumatic heart disease), problems with the heart (carditis), problems around pregnancy and birth (obstetric complications and foetal/neonatal events), death (mortality), whether people stuck to their treatment (treatment adherence), other problems such as dangerous breathing problems (anaphylaxis), complications such as nerve injury and whether the people included were happy with having antibiotics.

What did we find?

We found 11 studies (3951 participants) to help us answer our questions. People in these studies were an average of 12.3 years of age and were 50.6% male. Most had had previous episodes of rheumatic fever.

We found that using long-term antibiotics (either injected into the muscle every month or taken as a tablet every day) compared with no antibiotic probably reduces the risk of getting more episodes of rheumatic fever. The injection into a muscle route probably works better than the tablets. If you have the early stages of rheumatic heart disease picked up on an echocardiogram of the heart (a scan that uses sound waves to see the internal structure of the heart), then penicillin antibiotics injected into the muscle every month compared with no antibiotic likely reduces the risk of these heart problems getting worse. We found some evidence that antibiotics injected into the muscle compared with no antibiotics may not cause a very high risk of allergic reaction that affects breathing (anaphylaxis), but probably comes with a higher chance of redness at the injection side and allergic reactions to antibiotics. There was not much information on death rates or nerve injury, and no evidence on whether an antibiotic injection is better than tablets for preventing latent (early) rheumatic heart disease getting worse.

What are the limitations of the evidence?

The majority of the included studies (nine) were not carried out in low-income countries, which currently have the most cases of rheumatic heart disease. This makes these results potentially less relevant to people in these countries. There is also little information on important questions other than recurrence of rheumatic fever or progression of rheumatic heart disease. There were some other potential limitations in the evidence: we flagged six studies as having issues with blinding (study participants or staff knowing whether they received antibiotics and so potentially answering based on this information). It is possible that people in many of the included studies were aware of which treatment they were getting. Four studies may have had a problem with the process for placing people randomly into groups. For some of the results in the review, we only had one study.

More high-quality work is needed that is relevant to the parts of the world where rheumatic fever is currently most common. More research looking at early (latent) rheumatic heart disease, where the biggest differences may be made, is also needed.

How up-to-date is this evidence?

The evidence is current to 10 March 2024. Whilst this is the most up-to-date review available currently, most of the evidence in this review is from the 1950s to 1960s, so some of the treatments may be outdated.

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SUMMARY OF FINDINGS

Summary of findings 1. Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis

Antibiotic prophylaxis versus no antibiotic prophylaxis

Patients or population: participants with rheumatic fever or rheumatic heart disease

Settings: community and healthcare settings

Intervention: oral or intramuscular antibiotics

Comparison: standard medical therapy or placebo without antibiotic prophylaxis

Follow-up: longest available time point

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	NNTB (95% CI)	Notes
	Risk without antibiotics	Risk with antibiotics					
Recurrence of rheumatic fever (count data)	17 per 1000	7 per 1000	RR 0.39 (0.22 to 0.69)	1721 participants (6 trials)	⊕⊕⊕○ MODERATE[a]	NNTB 33 (25 to 100)	Defined using the Jones criteria or modified Jones criteria in three studies (Beaton 2022; Cope 1960; Feinstein 1966), and clinically in four studies (Brick 1950; Evans 1950; Gale 1952; Padmavati 1973)
Progression of rheumatic heart disease (latent)	81 per 1000	7 per 1000	RR 0.09 (0.03 to 0.29)	818 (1 trial)	⊕⊕⊕○ MODERATE[b]	NNTB 14 (10 to 20)	Defined by echocardiography according to World Heart Federation criteria
All-cause mortality (late-stage)	78 per 1000	96 per 1000	RR 1.23 (0.78 to 1.94)	994 (1 trial)	⊕⊕○○ LOW[c]	-	
Adverse events: anaphylaxis	0 per 1000	2 per 1000	Peto OR 7.39 (0.15 to 372)	818 (1 trial)	⊕⊕○○ LOW[d]	-	
Adverse events: sciatic nerve injury	0 per 1000	2 per 1000	Peto OR 7.39 (0.15 to 372)	818 (1 trial)	⊕⊕○○	-	

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					LOW[d]	
Adverse events: delayed hypersensitivity or allergic reaction	0 per 1000	152 per 1000	RR 137 (8.51 to 2210)	894 (2 trials)	⊕⊕⊕○ MODERATE[e]	NNTH 13 (-10 to 4)
Adverse events: local reactions to injection	0 per 1000	34 per 1000	RR 29 (1.74 to 485)	818 (1 trial)	⊕⊕⊕○ MODERATE[f]	NNTH 33 (50 to 20)

Note: further data are available in the main results for 'Recurrence of rheumatic fever (rate data)', 'Progression of rheumatic heart disease (late-stage)' and 'All-cause mortality (latent RHD)'.

*The risk in the intervention group (and its 95% CI) is calculated based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI), except where the assumed risk was found to be zero. In these cases, the corresponding risk was calculated in the same way as the assumed risk, according to event rates.

CI: confidence interval; **NNTB:** number needed to treat for an additional beneficial outcome; **NNTH:** number needed to treat for an additional harmful outcome; **№:** number; **OIS:** optimal information size; **OR:** odds ratio; **RHD:** rheumatic heart disease; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level for risk of bias (random sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting).
[b]Downgraded one level for imprecision. Risk of bias was not downgraded for this outcome as personnel were blinded.
[c]Downgraded one level for risk of bias (random sequence generation, allocation concealment, incomplete outcome data) and one level for imprecision.
[d]Downgraded two levels for imprecision, as results are based on one study with a very low event rate.
[e]Downgraded one level for imprecision, as results are based on events from one study.
[f]Downgraded one level for imprecision, as results are based on one study.

Summary of findings 2. Comparison 2 - Intramuscular penicillin versus oral antibiotic

Intramuscular penicillin versus oral antibiotic

Patients or population: participants with rheumatic fever or rheumatic heart disease

Settings: community and healthcare settings

Intervention: intramuscular penicillin

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Comparison: oral antibiotics

Follow-up: longest available time point

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	NNTB (95% CI)	Notes
	Risk with oral antibiotics	Risk with IM penicillin					
Recurrence of rheumatic fever (count data)	10 per 1000	1 per 1000	RR 0.07 (0.02 to 0.26)	395 (2 trials)	⊕⊕⊕◯ MODERATE[a]	NNTB 8 (95% CI 4 to 50)	Defined using the modified Jones criteria in Wood 1964, clinically in Markowitz 1957.
Progression of rheumatic heart disease (latent)	No data available						
All-cause mortality (late-stage)	7 per 1000	2 per 1000	Peto OR 0.22 (0.01 to 4.12)	431 (1 trial)	⊕◯◯◯ VERY LOW[b]	-	
Adverse events: anaphylaxis	No data available						
Adverse events: sciatic nerve injury	No data available						
Adverse events: delayed hypersensitivity or allergic reaction	No data available						
Adverse events: local reactions to injection	No data available						

Note: further data are available in the main results for 'Recurrence of rheumatic fever (rate data)', 'Progression of rheumatic heart disease (late-stage)' and 'All-cause mortality (latent)'.

*The risk in the intervention group (and its 95% CI) is calculated based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI), except where the assumed risk was found to be zero. In these cases, the corresponding risk was calculated in the same way as the assumed risk, according to event rates.

CI: confidence interval; **IM:** intramuscular; **NNTB:** number needed to treat for an additional beneficial outcome; **№:** number; **OR:** odds ratio; **RHD:** rheumatic heart disease; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

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[a]Downgraded one level for risk of bias (blinding, incomplete outcome data).
[b]Downgraded one level for risk of bias (blinding, incomplete outcome data), one level for imprecision and one level for indirectness, as RHD is a progressive disease and the effect on mortality is likely to be delayed, even beyond the endpoint of the study.

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BACKGROUND

Description of the condition

Rheumatic fever is an inflammatory disease, which develops after pharyngitis caused by the bacterium *Streptococcus pyogenes,* a gram-positive coccus. There is some evidence of a potential association between the development of rheumatic fever and group A streptococcal infection, including throat and skin infections in up to 3% to 5% of untreated individuals, and risk factors appear to include age, sex, environmental factors and access to antibiotics (Carapetis 2016; Oliver 2021; Siegel 1961). The condition manifests two to four weeks after the infection and is multisystemic, potentially involving the skin, joints, brain and heart. In rheumatic fever, carditis (i.e. inflammation in any layer of the heart: endocardium, myocardium or pericardium) can be observed. Over time, inflammation of the valve endothelium can occur repeatedly and lead to valvular damage and rheumatic heart disease (RHD) (Seckeler 2011; Zühlke 2015). This reaction, following one or more episodes of rheumatic fever, can result in fibrosis of any of the four heart valves, most commonly affecting the mitral valve (Seckeler 2011; Torres 2021; Zühlke 2015). The disease process can result in both stenotic or regurgitant valves, or both, routinely diagnosed on standard 2D transthoracic echocardiography. Valvular pathology is assessed against World Heart Federation (WHF) criteria to confirm or refute the diagnosis of valvular RHD (Reményi 2012; Zühlke 2015).

Left untreated, the chronic phase of the disease process can result in heart failure and ultimately death. It is the leading cause of cardiovascular morbidity and mortality in young people across the world, with prevalence highest in low-income countries. It causes 1.4 million deaths every year (Zühlke 2015). RHD remains the largest global cause of cardiovascular disease in the young (age < 25 years) (Ghamari 2022). RHD was the cause of 319,400 deaths worldwide in 2015 and was responsible for a loss of 10.5 million disability-adjusted life-years (Horton 2021). RHD largely affects those living in poor socioeconomic conditions with inadequate access to health care and with greater unchecked exposure to group A *Streptococcus*. Poor socioeconomic conditions likely lead to an increased propensity for the development of subsequent rheumatic fever and its associated long-term complications. The highest age-adjusted mortality due to RHD has been observed in Oceania, South Asia and central sub-Saharan Africa (Watkins 2017). Obstetric complications, such as chronic heart failure, acute heart failure admissions with arrhythmias, emergency surgery, such as percutaneous balloon mitral commissurotomy or surgical valve replacement, emergency delivery and death of the foetus or mother can occur in pregnant women with RHD (Cupido 2021).

Rheumatic fever and RHD can be seen as a spectrum of disease and categorised along this trajectory, as follows:

1. Medically diagnosed rheumatic fever with or without cardiac involvement - acute rheumatic fever is diagnosed using the modified Jones criteria (mJC) (Gewitz 2015).
2. Mild or latent RHD (detected by screening echocardiogram or active case finding) - RHD has an initial asymptomatic latent stage (latent RHD) that is detectable by echocardiography and is highly variable. This is defined using the 2012 WHF guideline (Reményi 2012). The term 'latent' is no longer recommended and has been superseded by two stage categories in the 2023 WHF guideline (Rwebembera 2023): mild echocardiographic criteria for RHD and mild RHD (stages A and B, respectively).
3. Clinically detected, established RHD (stages C and D of the 2023 WHF guideline) - if RHD is severe it may cause symptoms, prompting an individual to seek medical assessment, when a diagnosis of RHD can be made using clinical history that may indicate a previous episode of rheumatic fever and auscultation of a murmur. With progression of valvular damage, RHD may subsequently progress, causing heart failure and ultimately death (Watkins 2017).

There is a scarcity of therapeutic options for acute rheumatic fever and RHD. In the acute, inflammatory stage of rheumatic fever, there are no effective immunomodulatory treatments for carditis. In chronic RHD, access to specialist medical care and surgery is severely constrained in most low-socioeconomic, endemic regions of the globe. Hence, secondary antibiotic prophylaxis is the only treatment modality for many patients with acute rheumatic fever or RHD. The cost-effectiveness of secondary prophylaxis is well supported by data from multiple studies (Dixit 2023; Manji 2013; Reeves 2011; Ubels 2020).

Description of the intervention

Long-term antibiotic prophylaxis to prevent recurrence of rheumatic fever has been recommended because prophylaxis is thought to reduce the risk of recurrences of rheumatic fever. Moreover, antibiotic prophylaxis is relatively cost-effective (Dixit 2023). Duration of antibiotic prophylaxis is currently based on the severity of heart disease (Gerber 2009; Zühlke 2013). Where possible, patients should be registered with a regional rheumatic fever prevention programme to facilitate long-term follow-up (Ralph 2021; Torres 2021).

Secondary prevention is defined as the continuous administration of antibiotics to a patient with a previous episode of rheumatic fever in an attempt to reduce recurrent episodes. The cornerstone of contemporary secondary prophylaxis is intramuscular benzathine benzylpenicillin every three weeks (or every four weeks in low-risk areas or patients), 1,200,000 units if > 30 kg and 600,000 in children < 30 kg. Oral penicillin is considered an alternative, although this is considered less efficacious (penicillin V, 250 mg twice daily). Oral macrolides (erythromycin, 250 mg twice daily) or sulfadiazine or sulfisoxazole (1 g if > 30 kg and 500 mg if < 30 kg) can be considered in penicillin allergy (Cilliers 2006; Ralph 2021; WHO Expert Consultation 2001).

How the intervention might work

Once valvular RHD has become advanced and symptomatic, treatment is generally focused on corrective treatment of valvular disease and medical management of congestive cardiac failure and other complications like atrial fibrillation (Negi 2021). It is thought that secondary antibiotic prophylaxis started in early sub-clinical phases of RHD may reduce progression to clinical or advanced RHD by reducing the number of asymptomatic or subclinical colonisation events by group A *Streptococcus*, which may trigger the worsening or re-initiation of the self-immunity via a type II hypersensitivity reaction, and thus inflammatory damage to cardiac structures, including valves (Jaiteh 2021). It is hypothesised that secondary prophylaxis in severe RHD prevents damage and progression to the affected valve as well as protecting unaffected valves. Systematic echocardiography was first shown to

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be superior to auscultation in screening for RHD by *Marijon and colleagues* (Marijon 2007), and is now widely accepted (Beaton 2012; Roberts 2013). Identification of these initial, sub-clinical and silent rheumatic valve lesions could constitute an opportunity for early intervention (Minozzi 2022).

Why it is important to do this review

It has not yet been conclusively proven that early RHD is treatable with penicillin therapy or that early antibiotic treatment alters long-term progression of the disease. This has led to uncertainty as to whether all patients diagnosed with rheumatic fever or early signs of RHD (or both) on cardiac ultrasound should be given long-term antibiotic treatment for secondary prevention of rheumatic fever recurrence and progression to advanced RHD (Beaton 2017). However, patients on regular secondary prophylaxis have been observed to obtain partial or total recovery from valve lesions, and some of them have been discharged without echocardiographic abnormalities (Torres 2021). Moreover, secondary prophylaxis programmes appear safe and can be successfully conducted (Spinetto 2011). Additionally, the best choice of antibiotics for secondary prevention of rheumatic fever is still a matter of debate (Beaton 2019). Evidence in the field has faced some criticism, as it is composed of longitudinal studies with issues observed for ensuring effective secondary prophylaxis delivery and the completeness of reporting and monitoring of the progression of RHD. Potential difficulties for the success of secondary antibiotic prophylaxis include shortage of penicillin and other antimicrobials worldwide; a lack of comprehensive community-based service delivery systems to treat at-risk individuals; and, whilst there is not yet specific evidence of resistance of group A *Streptococcus* to penicillins, there is a responsibility of antibiotic stewardship for the user, necessitating a balanced use in the context of increasing global rates of resistance (Katzenellenbogen 2020; Klein 2018; Marijon 2021; Ralph 2016; Rémond 2016; Torres 2021). Conclusions drawn from this review will be hugely important to inform clinical practice guidelines, including World Health Organization (WHO) and national guidelines.

OBJECTIVES

1. To assess the effects of long-term antibiotics versus no antibiotics (control) for secondary prevention of rheumatic fever recurrence and associated sequelae in people with previous rheumatic fever or RHD.
2. To assess the effects of long-term intramuscular (IM) penicillin versus long-term oral antibiotics for secondary prevention of rheumatic fever recurrence and associated sequelae in people with previous rheumatic fever or RHD.

METHODS

Criteria for considering studies for this review

The protocol for this review was published a priori (Pelone 2022).

Types of studies

We included randomised controlled trials (RCTs) or quasi-randomised trials with individual or cluster allocation. Quasi-randomised trials were defined as trials where allocation had been made by some quasi-random method of allocation, such as alternation or date of birth. We included studies in any language.

We excluded the following.

1. Non-randomised studies, as we believe enough RCT evidence on this topic is available, and the methodology of available non-randomised studies would be of lower quality and likely to introduce potential biases into the review.
2. Cross-over trials.
3. Publications that were not original research (e.g. reviews, editorials and letters).
4. Theses and book chapters.
5. Animal or laboratory studies not carried out in a clinical setting.
6. Purely epidemiological reports (i.e. only demographics and mortality rate, with no clinical characteristics).
7. Case reports and series describing rheumatic fever or RHD patients.
8. Longitudinal and cross-sectional survey research.
9. Studies conducted prior to 1950, as currently used antibiotics were not in common usage prior to this date.

Types of participants

Participants diagnosed with rheumatic fever or RHD, of all age groups, in community or hospital settings. The criteria generally used for diagnosing these conditions are the WHF criteria for diagnosing RHD (Reményi 2012) and the Jones criteria for rheumatic fever (Gewitz 2015). We have used the definition for diagnosis given in each paper. We have included studies in which more than 80% of participants had a previous diagnosis of rheumatic fever or RHD.

Types of interventions

Comparison 1

Antibiotics versus no antibiotic prophylaxis.

Antibiotics were defined as the following, clinically relevant regimes:

1. oral benzylpenicillin (penicillin G) (50,000 units to 6 million units, daily)/intramuscular benzathine benzylpenicillin (BPG) (600,000 units to 1.2 million units, administered three to four times weekly);
2. phenoxymethylpenicillin (penicillin V) (250 mg to 1 g, twice to four times daily);
3. sulfadiazine (500 mg to 1 g daily);
4. macrolide or azalide antibiotic (e.g. 250 mg to 500 mg, twice daily); or
5. any other antibiotics and any form of administration (these could have been in any form, i.e. intramuscular, intravenous or oral, and have been given at any frequency or for any length of course).

We included studies with an antibiotic prophylaxis duration of at least six months.

The comparator was standard medical therapy or placebo without antibiotic prophylaxis.

We considered medications other than antibiotic prophylaxis eligible as concomitant medications, provided they were applied to all treatment arms.

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Comparison 2

Intramuscular penicillin versus oral antibiotics.

The comparator was oral antibiotics.

Other comparisons

There are several potential comparisons that were outside the scope of this review, but are listed here for clarity:

1. secondary prophylaxis, comparing excellent versus suboptimal adherence to intramuscular or oral antibiotics;
2. oral prophylaxis - type of oral antibiotic (non-penicillin versus penicillin-based regimen); and
3. three- versus four-weekly BPG regimens.

The impact of local anaesthetic to reduce injection pain and improve treatment adherence has been assessed by a separate review for the WHO guideline (Pelone 2024).

Types of outcome measures

We extracted data on the following outcomes at the longest period of follow-up (defined as six months or longer). The outcomes were not used as an eligibility criterion.

Primary outcomes

1. Recurrence of rheumatic fever (preferentially diagnosed using the Jones criteria, otherwise as per trial protocol).
2. Progression or severity of RHD (defined as per trial protocol).
3. Cardiac complications (e.g. arrhythmia, valvular heart disease, endocarditis, heart failure) (defined as per trial protocol).

Secondary outcomes

1. Obstetric complications (defined as per trial protocol).
   a. Maternal events, for example:
      i. death of mother
      ii. heart failure admission;
      iii. need for cardiac intervention;
      iv. percutaneous balloon mitral commissurotomy; or
      v. surgical valve replacement.
   b. Foetal/neonatal events (defined as per trial protocol), for example:
      i. foetal death;
      ii. neonatal death (< 1 week and > 1 week of age);
      iii. miscarriage;
      iv. low Apgar score (Cnattingius 2020);
      v. preterm birth; or
      vi. small for gestational age.
2. Mortality (defined as per trial protocol).
3. Treatment adherence (defined as per trial protocol).
4. Adverse events (any) (defined as per trial protocol).
5. Acceptability to participants (defined as per trial protocol).

Search methods for identification of trials

Electronic searches

We searched the following sources from inception up to 10 March 2024.

1. Cochrane Central Register of Controlled Trials (CENTRAL; 2022, Issue 7) in the Cochrane Library (searched 10 March 2024).
2. MEDLINE Ovid SP (1946 to 10 March 2024).
3. Embase Ovid SP (1974 to 10 March 2024).
4. Conference Proceedings Citation Index-Science Web of Science (CPCI-S; 1990 to 11 March 2024).
5. US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov; searched 10 March 2024).
6. World Health Organization International Clinical Trials Registry Platform (WHO ICTRP; apps.who.int/trialsearch; searched 10 March 2024).
7. International Standard Randomised Controlled Trial Number (ISRCTN; www.isrctn.com; searched 10 March 2024).

Search strategies were developed by consulting clinicians, controlled vocabularies (e.g. MeSH and Emtree), literature reviews and testing search results. Based on the recommendations in the *Cochrane Handbook for Systematic Reviews of Interventions* (Lefebvre 2022), searches balanced sensitivity with specificity, without applying methodological search filters. The search was not limited by publication date, language, status or document type. Final versions of the search strategies were peer-reviewed by an information specialist (search strategies are available in Appendix 1).

Searching other resources

We checked the reference lists of the included studies and other relevant systematic reviews identified for additional references to trials. In addition, we examined any relevant retraction statements and errata for included studies.

Data collection and analysis

Selection of studies

Six review authors (JJHB, SAA, MS, ST, MA, JY) independently screened the titles and abstracts of all records yielded by the search against our inclusion criteria and classified them as eligible or not eligible, using Rayyan software (Ouzzani 2016); JJHB and MA screened records from the updated search performed on 10 March 2024. After retrieval of full-text papers, six review authors (JJHB, SAA, MS, ST, MA, JY) independently screened the full texts and identified studies for inclusion and exclusion, with each full-text paper reviewed by at least two review authors working independently. Disagreements were resolved through the involvement of a third review author (JJHB, MA) at both stages of the review process. Whilst these authors were involved in the initial screen, there were no disagreements between JJHB and MA, hence either author added a new, impartial opinion. We identified and excluded duplicates and collated multiple reports of the same study so that each study, rather than each report, was the unit of interest in the review. We recorded and reported the reasons for exclusion of full-text reports of studies expected to be in the review. We completed a PRISMA flow diagram (Liberati 2009) and a Characteristics of excluded studies table. Where texts were written in languages other than English, we used Google Translate to assess them for eligibility (Google Translate).

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Data extraction and management

We sought to include the following data, where available. We recorded the data on a data collection form, which was initially piloted by two authors (MA, JJHB).

1. Methods: trial design, total trial duration, number of trial centres, trial location, trial setting and date of the trial.
2. Participants: number (n) of participants, sex, average age, age range, inclusion criteria, type of housing, number (n) of persons living in the household, number (n) of persons < 15 living in the household, type of school, water/sanitation, assets, maternal education, income (WAMI) index, average duration of maternal education, sore throat reported in < 4 weeks, skin infection reported < 4 weeks, at least one first-degree family relative with a previous diagnosis of rheumatic fever, at least one first-degree family relative with a previous diagnosis of RHD, previous acute rheumatic fever, history of RHD, WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of the trial and other comorbidities such as NYHA (New York Heart Association) class at baseline and end of the trial.
3. Interventions: form, dosage and type of antibiotics, length of antibiotic course.
4. Comparator: form, dosage, type and length of comparator (placebo, standard medical therapy or oral antibiotics).
5. Outcomes: outcomes specified, including definitions, time points and methods for collection.
6. Notes: funding for trials, notable conflicts of interest of trial authors and power calculations.

Four review authors (JJHB, ST, JY, SAA) independently extracted outcome data from each included study into a pre-formatted Excel spreadsheet, with at least two review authors independently checking the extracted data against the original publication. Any disagreements were resolved by consensus or by involving a third review author (MA). One review author (JJHB) transferred data into the Review Manager (RevMan) file (RevMan 2022). We double-checked that data had been entered correctly by comparing the data presented in the systematic review with the data extraction form. A second check of extracted data was performed by review authors (JJHB, MA). We contacted the authors of studies on an as-required basis to obtain data or information.

Assessment of risk of bias in included studies

We evaluated the risk of bias in the included studies using the original Cochrane risk of bias tool (RoB 1) (Higgins 2011a).

At least two review authors (MA, ST, HG, JJHB) independently assessed risk of bias for each study using RoB 1, as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011a). We resolved any disagreements by discussion or by involving another author (either JJHB or MA), not involved in the original assessment. We assessed the risk of bias across the following domains.

1. **Random sequence generation** - studies were considered 'low' risk if there was evidence of any truly random process (e.g. computer random number generator) or 'high' risk if there was a non-random component in the sequence generation process.
2. **Allocation concealment** - studies that described adequate allocation concealment in which participants and investigators could not foresee assignment were considered 'low' risk, whereas studies that did not describe such methods were considered 'high' risk.
3. **Blinding of participants and personnel** - if participants and personnel had no knowledge of the allocated interventions, then a study was considered 'low' risk, and 'high' risk if they were or could have been aware.
4. **Blinding of outcome assessment** - a study was considered 'low' risk of bias if outcome assessors had no knowledge of the allocated interventions or the outcome measurement was not likely to be influenced by lack of blinding, whereas a study was considered 'high' risk if there was no blinding, or it was likely that blinding could have been broken and the outcome could have been influenced by a lack of blinding.
5. **Incomplete outcome data** - if there was a low risk of attrition bias, for example if there were no missing outcome data or missing outcome data were balanced across intervention groups, then a study was considered 'low' risk. If there were features consistent with a high risk of attrition bias, for example missing outcome data likely related to true outcome, then a study was considered 'high' risk.
6. **Selective outcome reporting** - a study was considered 'low' risk if all pre-specified outcomes were reported as documented in a study protocol or if there was no protocol but the published report includes all expected outcomes, whereas a study was considered 'high' risk if there were significant deviations in terms of outcomes or measurements from a pre-specified protocol or if an outcome was reported incompletely.
7. **Other bias** - we sought 'high-risk' features of other bias including potential sources of bias relating to specific study design, fraudulent claims or other problems. A study was considered 'low' risk if it appeared to be free of other sources of bias.

In all categories, if there was insufficient information to deem a study either 'low' or 'high' risk, then the study was considered 'unclear' risk. We provided direct quotes from the study report, with justifications for our judgement in the risk of bias tables in the Characteristics of included studies table. Where information on risk of bias related to unpublished data or correspondence with trialists, we noted this in the risk of bias tables.

We were interested in quantifying the effect of assignment to the interventions at baseline, ideally with the interventions received as intended (the 'intention-to-treat effect'), but this was not considered essential.

Measures of treatment effect

The data we collected were dichotomous and were analysed as risk ratios (RR) with 95% confidence intervals (CI). We analysed rate data as rate ratios with 95% CI (Deeks 2011). An exception to this was outcomes with very low event rates (< 1%). For such cases, we used the Peto one-step odds ratio (OR) method, as recommended by the *Cochrane Handbook* (Higgins 2011b).

Unit of analysis issues

Multi-arm studies

When we identified multi-arm studies (e.g. two arms with different antibiotics and a control arm), we combined the antibiotic arms for comparison 1 (antibiotics versus control) and separated the arms

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for comparison 2 (intramuscular antibiotics versus oral antibiotics), and conducted a subgroup analysis, as required.

Multiple follow-up

For studies that reported on more than one follow-up, we analysed outcomes at the longest possible time of follow-up. It was not possible to analyse outcomes at different time points as separate comparisons to avoid a unit of analysis error.

Dealing with missing data

It was not possible to contact investigators or trial sponsors in cases of missing data due to the age of the studies. The missing data for participants or summary data were assumed to be missing at random and we explored the impact of excluding such studies in the overall assessment of results by a sensitivity analysis. We did not use last observation carried forward or make assumptions of particular values such as best-case scenarios or worst-case scenarios.

Assessment of heterogeneity

We assessed studies for clinical and methodology heterogeneity (we report the results of this assessment in the 'Description of studies' and 'Risk of bias in included studies' sections).

We inspected forest plots visually, to consider the direction and magnitude of effects and the degree of overlap between confidence intervals. We used the I^2 and tau^2 statistics to measure statistical heterogeneity among the studies in each analysis. We acknowledge that there is substantial uncertainty in the value of I^2 when there is only a small number of studies. We also considered the P value from the Chi^2 test.

Where we identified substantial heterogeneity, we reported it, and explored possible causes by prespecified subgroup analysis. We considered heterogeneity as substantial if there was a low P value (less than 0.10) in the Chi^2 test for heterogeneity, or if tau^2 was greater than zero. Strict thresholds for interpreting I^2 are not recommended and we therefore followed the rough guide as outlined in the *Cochrane Handbook for Systematic Reviews of Interventions*, where 50% or more represents substantial to considerable heterogeneity (Deeks 2022). We used the Chi^2 test provided in RevMan (RevMan 2022) to compare differences in heterogeneity between subgroups, where relevant (Deeks 2022).

Assessment of reporting biases

We obtained and assessed trial protocols, when possible, for selective reporting as part of the risk of bias assessment.

Data synthesis

We performed a meta-analysis where outcomes, participants, interventions and comparisons were judged sufficiently similar, ensuring the answer was sufficiently clinically meaningful for pooling to make sense. Where this was not possible or deemed inappropriate, we described the information narratively.

We used a random-effects model due to the high probability of heterogeneity in the RCTs and other evidence included in this review, along with the commonly used and validated Mantel-Haenszel method. We used a fixed-effect model with the Peto method, as recommended in Chapter 10 of the *Cochrane Handbook* (Deeks 2022).

Subgroup analysis and investigation of heterogeneity

We carried out subgroup analyses for the factors listed below for all outcomes, but only for our primary time point of interest. Our time point of interest was the longest follow-up available.

1. Rheumatic fever with carditis.
2. Severe valvular RHD.
3. Age stratification (children < 5 years, 5 to 15 years, 16 to 17 years, > 18 years, adults).
4. Duration of treatment (< 5 years, 5 to 9 years, > 10 years, up to a certain age and life-long).
5. Class of antibiotic.

We used the formal test for subgroup differences in RevMan 2022 on which to base our interpretation.

Where dichotomous subgroup analysis was not appropriate, we used meta-regression. This was performed in Stata 17 using a random-effects restricted maximum likelihood (REML) model meta-analysis, followed by meta-regression with the subgroup factor of interest (Stata). The results from this analysis are included as a figure.

Sensitivity analysis

All studies were included in the primary analysis. To assess the potential effects of studies at high risk or high risk/some concerns, we carried out sensitivity analyses. We conducted sensitivity analyses for:

- low risk of bias studies;
- peer-reviewed publications;
- missing data;
- per-protocol analysis;
- exclusion of quasi-randomised trials;
- oral benzylpenicillin;
- diagnosis of rheumatic fever; and
- contemporary trials.

It was noted that three studies used oral benzylpenicillin (Cope 1960; Feinstein 1966; Markowitz 1957). As this formulation of penicillin is now rarely used orally, we applied a sensitivity analysis to investigate the exclusion of these studies.

Summary of findings and assessment of the certainty of the evidence

Using GRADEpro GDT software (GRADEpro GDT), we produced summary of findings tables for each of our comparisons:

1. antibiotic prophylaxis versus no antibiotic prophylaxis; and
2. intramuscular penicillin versus oral antibiotic.

We conducted the second comparison irrespective of whether there was evidence for the first.

We included the following, most clinically relevant outcomes in the summary of findings tables:

- recurrence of rheumatic fever;
- progression or severity of rheumatic heart disease;
- mortality; and

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- severe side effects (e.g. anaphylaxis).

We included data from the longest available follow-up time point.

GRADE assessment

We used the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness and publication bias) to assess the certainty of the body of evidence as it related to the studies that contributed data to the meta-analyses for the prespecified outcomes included in the summary of findings tables (Schünemann 2013). We used the overall risk of bias judgement from the Cochrane risk of bias tool as part of the GRADE assessment for each outcome (Schünemann 2023). This corresponded to a summary rating for each outcome of: 'very low' certainty, 'low' certainty, 'moderate' certainty and 'high' certainty. Corresponding risk was calculated as assumed risk multiplied by the relative effect estimate, except where the assumed risk was zero, where in this case the corresponding risk was calculated directly from the event rates in the same way as assumed risk, as the former method would be misleading (Schünemann 2011). To investigate imprecision, we calculated the optimal information size (OIS) using default type I error probability and power (α 0.05, power 0.80), by comparing proportions of two independent samples, as recommended in the *GRADE Handbook* (GRADEpro GDT).

Judgements about the certainty of the evidence were made by two review authors working independently (JJHB, MA), with disagreements resolved by discussion or involving a third author (RP). We justified all decisions to downgrade the certainty of the evidence (to a maximum of three levels) using footnotes and added comments to aid the reader's understanding of the review, where necessary. We incorporated the ratings into the reporting of results for each outcome.

We extracted study data, formatted our comparisons in data tables and prepared a summary of findings table before writing the results and conclusions of our review.

RESULTS

Description of studies

Please see Characteristics of included studies, Characteristics of excluded studies and Characteristics of ongoing studies.

Results of the search

We identified 3236 records from our systematic search and six records from other sources, including backward and forward citation searches and peer review. We removed 1016 duplicates and screened 2226 records. Many of the screened studies were relatively old, dating back to the 1950s, and were not available with digital abstracts. For 21 records, we sought photocopies of the original paper copies of the articles from the British Library, as there was insufficient information to make a screening assessment, and in several cases we had to translate titles and abstracts into English using Google Translate (Google Translate) before we could make a judgement. We excluded 2178 records as irrelevant and brought forward 48 reports for full-text assessment. We excluded 20 studies and included 11 eligible studies (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Markowitz 1957; Padmavati 1973; Wood 1964). There was one ongoing study (NCT05693545). See Figure 1.

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Figure 1. PRISMA flow diagram.

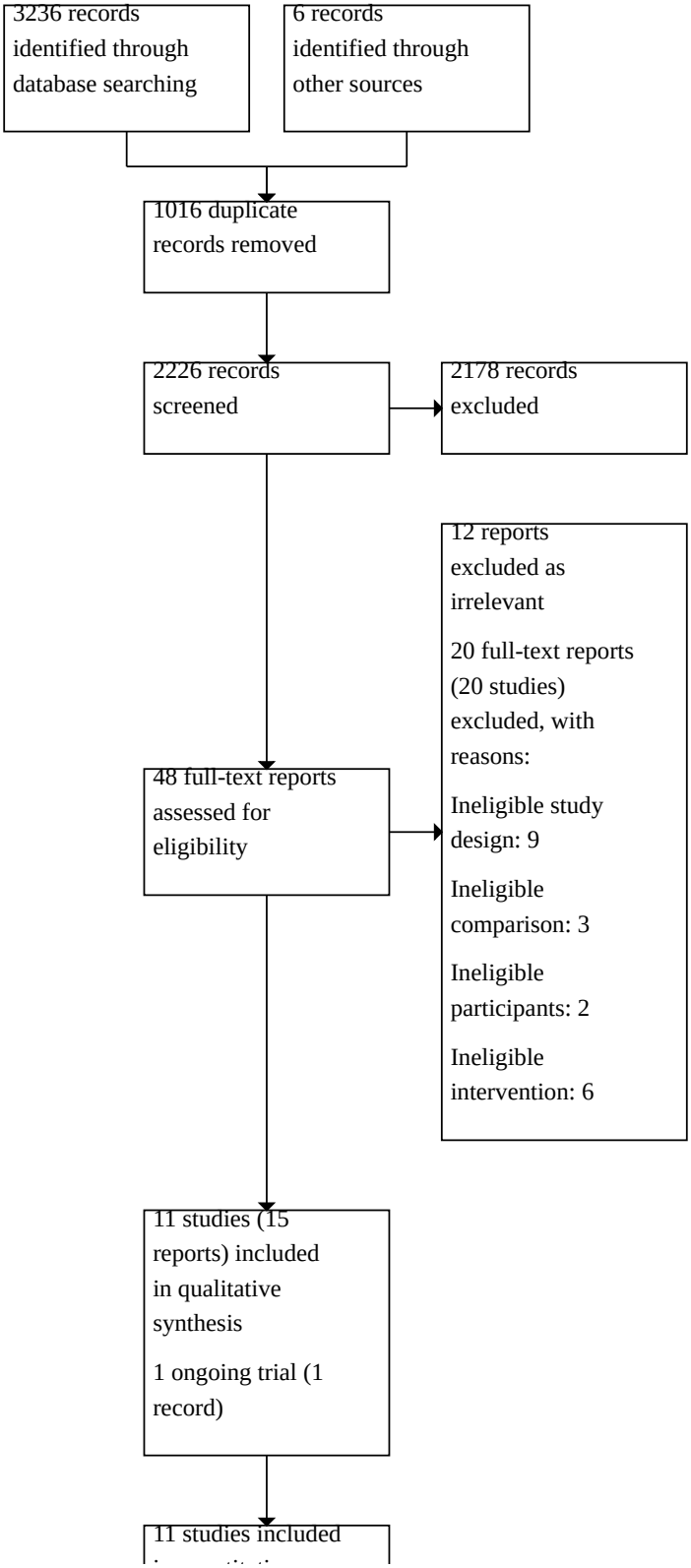

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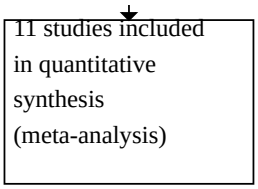
Figure 1. (Continued)



Two eligible studies were identified from the previous Cochrane review in this area (Manyemba 2002), and these were generally older records (Feinstein 1959; Feinstein 1968), and one additional study (Brick 1950) was identified from the introduction of Gale 1952. One study was identified at peer review (Markowitz 1957). Manyemba 2002 included Feinstein 1965; however, we considered this as a duplicate due to identical methods, authorship and overlap in recruitment times with Feinstein 1968. It was not possible to contact study authors or sponsors due to the age of the manuscript. Manyemba 2002 did not include Brick 1950 and Padmavati 1973, labelling the studies as 'non-randomised' and relying on 'historical controls'. However, in our assessment, Padmavati 1973 allocated participants based on the days of the week and Brick 1950 alternated allocation to intervention and control; thus, we considered these trials to be quasi-randomised.

We identified additional reports for the following studies, which can be found under the 'references' section of these included studies: Beaton 2022; Feinstein 1959; Feinstein 1968; Wood 1964.

Included studies

Eleven studies met the inclusion criteria and included 3951 (standard deviation ± 323) participants (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957 Padmavati 1973; Wood 1964).

Setting

Nine included studies were run in high-income countries but during the 1950s to 1960s, with five from the United States of America (Feinstein 1959; Feinstein 1966; Feinstein 1968; Markowitz 1957; Wood 1964), three in the UK (Cope 1960; Evans 1950; Gale 1952) and one in Canada (Brick 1950). Two more recent studies were run between 1966 and 1971 and 2018 to 2020 and were conducted in India and Uganda, respectively (Padmavati 1973; Beaton 2022).

Eight studies took place in an outpatient setting (Brick 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Padmavati 1973; Wood 1964), two in an inpatient or dedicated convalescent home setting (Cope 1960; Evans 1950), and one was run in primary and secondary schools (Beaton 2022). The median duration of follow-up was 24 months and ranged from eight (Gale 1952) to 72 (Wood 1964) months.

Participants

Four studies included individuals with previous rheumatic fever as diagnosed by the mJC (Feinstein 1959; Feinstein 1966; Feinstein 1968; Wood 1964), two studies included participants with rheumatic fever diagnosed by mJC or clinical consensus (Cope 1960; Padmavati 1973), four trials included participants with previous rheumatic fever diagnosed without the Jones criteria (Brick 1950; Gale 1952; Evans 1950; Markowitz 1957), and one study included latent RHD diagnosed by echocardiography according to the 2012 World Heart Federation criteria (Reményi 2012), and this study excluded individuals with moderate or severe RHD (Beaton 2022). The sample sizes ranged from 73 (Gale 1952) to 994 (Padmavati 1973) (median 143). Individuals with moderate or severe RHD were excluded from Beaton 2022. Three studies that recruited participants with previous rheumatic fever specified that it had to have been within the range of six months to five years ago (Cope 1960; Evans 1950; Feinstein 1966). Seven included studies specifically recruited children (Beaton 2022; Brick 1950; Evans 1950; Feinstein 1959; Feinstein 1968; Gale 1952; Wood 1964), two recruited predominantly children (Cope 1960; Padmavati 1973), and one recruited participants aged > 14 years (Feinstein 1966). The mean age of participants was 12.3 (± 2.51) years (Beaton 2022; Brick 1950; Feinstein 1966; Feinstein 1968; Markowitz 1957; Padmavati 1973; Wood 1964). Three studies reported the sex of included participants as 49.4% female (Beaton 2022; Markowitz 1957; Padmavati 1973). Two studies had high rates of semi-permanent residency (Beaton 2022) and crowded living conditions with "questionable nutrition" (Padmavati 1973). Two studies commented upon schooling, with Beaton 2022 recruiting predominantly day schooled children, and Padmavati 1973 investigating predominantly children studying in schools in New Delhi. Padmavati 1973 investigated participants from families with predominantly low income, and Beaton 2022 reported an income WAMI index of 0.3 (± 0.1). Recent sore throats were reported in 18% to 42% of participants (Beaton 2022; Padmavati 1973), and one study reported an average of approximately four sore throats in both control and with antibiotics per participant during the study (Brick 1950). Beaton 2022 reported an average of 7.9 persons living in each household of participants, of which approximately half were below the age of 15 years, the average duration of maternal education was 5 years, and 1.3% and 2.3% had at least one first-degree relative with previous rheumatic fever and RHD, respectively. No information was available on water and sanitation, or maternal education. Three studies reported similar rates of group A *Streptococcus* between intervention and control groups of between 3% and 29% (Brick 1950; Gale 1952; Padmavati 1973). Two studies reported attacks of carditis as part of previous rheumatic fever in 15.6% (Padmavati 1973) and 37.9% (Feinstein 1966), Beaton 2022 did not recruit any participants with previous carditis by definition, and no other study commented on previous attacks of carditis.

When collecting data from Beaton 2022 and Padmavati 1973 on the progression of RHD, it became apparent that these two studies described two very different populations. Beaton 2022 concerned latent RHD and Padmavati 1973 included participants with 79.9%

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pre-existent chronic valvular heart disease; therefore, they were considered separately as latent and late-stage RHD, respectively.

Interventions

Of the seven studies meeting the inclusion criteria for comparison one (antibiotics versus no antibiotics), two trials used intramuscular benzathine benzylpenicillin once monthly (Beaton 2022: 0.6 to 1.2 million units; Padmavati 1973: no dose specified) and five used oral penicillin of between 50,000 to 5.25 million units (Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952). Of these five, one specified oral antibiotics as benzylpenicillin (Feinstein 1966) and one as benzylpenicillin and phenoxymethylpenicillin (Cope 1960). All these trials either appeared to give antibiotics for the duration of the trial (Brick 1950; Feinstein 1966; Gale 1952; Padmavati 1973) or specified that antibiotics were given for the duration of the trial (Beaton 2022; Cope 1960; Evans 1950).

All four studies meeting the inclusion criteria for comparison two (intramuscular versus oral antibiotics) gave intramuscular benzathine benzylpenicillin at 1.2 million units every month for between three and six years, which equated to the duration of each trial (Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964).

Antibiotics are used for a variety of reasons other than RHD prophylaxis in normal clinical practice. One trial reported giving antibiotics to either intervention or control for use in medical conditions other than rheumatic fever or RHD as necessary (Padmavati 1973). Even though this could act as a confounding factor if antibiotics were given to individuals within the control group, this was deemed acceptable for inclusion as it was likely that other included studies did not prevent individuals in the control group from receiving antibiotics for reasons other than RHD prophylaxis. Moreover, it would not be ethical to prevent individuals from using antibiotics for reasons other than RHD prophylaxis if they needed them.

Comparisons

Seven studies met the inclusion criteria for comparison one (oral or intramuscular versus no antibiotics). Five studies investigated antibiotics versus standard care without antibiotics (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Padmavati 1973), and two studies compared with placebo (Feinstein 1966; Gale 1952). Two studies used the intramuscular route (Beaton 2022; Padmavati 1973) and five studies used the oral route (Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952). Benzylpenicillin (Penicillin G) was used by four studies (Beaton 2022; Cope 1960; Feinstein 1966; Padmavati 1973), phenoxymethylpenicillin (penicillin V) by one study (Cope 1960), and three studies did not specify the type of penicillin used (Brick 1950; Evans 1950; Gale 1952).

Four studies met the inclusion criteria for comparison two (intramuscular antibiotics versus oral antibiotics) (Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964); Feinstein 1959 and Markowitz 1957 compared intramuscular benzathine benzylpenicillin with oral benzylpenicillin, Feinstein 1968 compared intramuscular benzathine benzylpenicillin with oral sulfadiazine, and Wood 1964 compared intramuscular benzathine benzylpenicillin with oral sulfadiazine and oral benzylpenicillin. Of the four studies in comparison two, one used 400,000 units of oral benzylpenicillin three times per day (Feinstein 1968), three used 200,000 units once daily (Feinstein 1959; Markowitz 1957; Wood 1964) and two included oral 1 g sulphasalazine daily (Feinstein 1959; Wood 1964).

Outcomes

There were sufficient data for the primary outcome in both comparisons (comparison one: Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973; comparison two: Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964), with a reasonable amount of data available for most secondary outcomes found predominantly in Beaton 2022 and Padmavati 1973 in comparison one and in Wood 1964 in comparison two. However, there were no data for obstetric complications (maternal or foetal events). Delayed hypersensitivity or allergic reaction was defined by Brick 1950 as "local reactions", and Beaton 2022 did not define this outcome beyond "minor allergic and hypersensitivity reactions".

The table below shows which included study reported data on each of our pre-defined primary and secondary outcomes, and at what time points.

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Outcome	Recurrence of rheumatic fever	Progres-sion/ sever-ity of RHD	Cardiac complica-tions	Obstetric complica-tions	Mortality	Treatment adherence	Adverse events	Acceptabil-ity
Beaton 2022	-	2 years	-	-	2 years	-	2 years	-
Brick 1950	2 years	-	-	-	-	-	2 years	-
Cope 1960	1 year	-	-	-	-	-	-	-
Evans 1950	23 months	-	-	-	-	-	-	-
Feinstein 1959	3 years	-	-	-	-	-	-	-
Feinstein 1966	28 months	-	-	-	-	-	-	-
Feinstein 1968	4 years	-	-	-	-	-	4 years	4 years
Gale 1952	8 months	-	-	-	-	-	-	-
Markowitz 1957	14 months	-	-	-	-	-	14 months	-
Padmavati 1973	5 years	5 years	-	-	5 years	-	-	-
Wood 1964	6 years	-	-	-	6 years	6 years	-	-

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Recurrence of rheumatic fever was defined using the Jones criteria or modified Jones criteria in six studies (Beaton 2022; Cope 1960; Feinstein 1959; Feinstein 1966; Feinstein 1968; Wood 1964), and clinically in five studies (Brick 1950; Evans 1950; Gale 1952; Markowitz 1957; Padmavati 1973). Progression of RHD was defined by echocardiography according to World Heart Federation criteria by Beaton 2022 and by clinical determination of heart disease in Padmavati 1973. Mortality was defined in all included studies as death during the trial's duration. Adverse events including anaphylaxis and sciatic nerve injury were defined according to clinical guidelines.

Trial design

Four studies were considered to be quasi-randomised (Brick 1950; Cope 1960; Feinstein 1966; Padmavati 1973) and the remaining seven studies were considered RCTs (Beaton 2022; Evans 1950; Feinstein 1959; Feinstein 1968; Gale 1952; Markowitz 1957; Wood 1964). One study had multiple treatment arms, including intramuscular benzylpenicillin versus oral penicillin versus oral sulfadiazine (Wood 1964, also see Stollerman 1955).

Funding sources

Funding sources were available for seven of the 11 included studies and consisted of grants and funding from charities in the majority of studies. Three studies were funded by charities (Beaton 2022; Gale 1952; Markowitz 1957) and four by government grants (Feinstein 1959; Feinstein 1966; Feinstein 1968; Wood 1964).

Excluded studies

We excluded 37 reports, including 19 for a lack of comparison, eight due to the ineligible trial design, four due to an ineligible comparison, two for no original data, two for being non-randomised, one for having ineligible participants and one for being an ongoing trial (Figure 1). We list 20 of these (those studies that appeared to meet the inclusion criteria, but only on closer inspection did not) in the 'excluded studies' table (Characteristics of excluded studies). Some examples include exclusion for being non-randomised (e.g. Spinetto 2011; Stollerman 1955; Tompkins 1972), or recruiting an ineligible population (e.g. recruiting pharyngitis as opposed to rheumatic fever or RHD; Chamovitz 1954).

Ongoing studies

One ongoing study that would likely meet the inclusion criteria was identified. It is titled the 'GOALIE' trial and is being conducted by Beaton and colleagues. It is a phase II trial investigating intramuscular benzylpenicillin versus phenoxymethylpenicillin in children aged 5 to 17 years with latent RHD. It looks to measure, in particular, RHD progression over a two-year duration (NCT05693545).

Risk of bias in included studies

The full results of our risk of bias assessment for each study can be found in the risk of bias tables under Characteristics of included studies. Below, we provide a summary of our results. See also the risk of bias graph (Figure 2) and risk of bias summary (Figure 3).

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

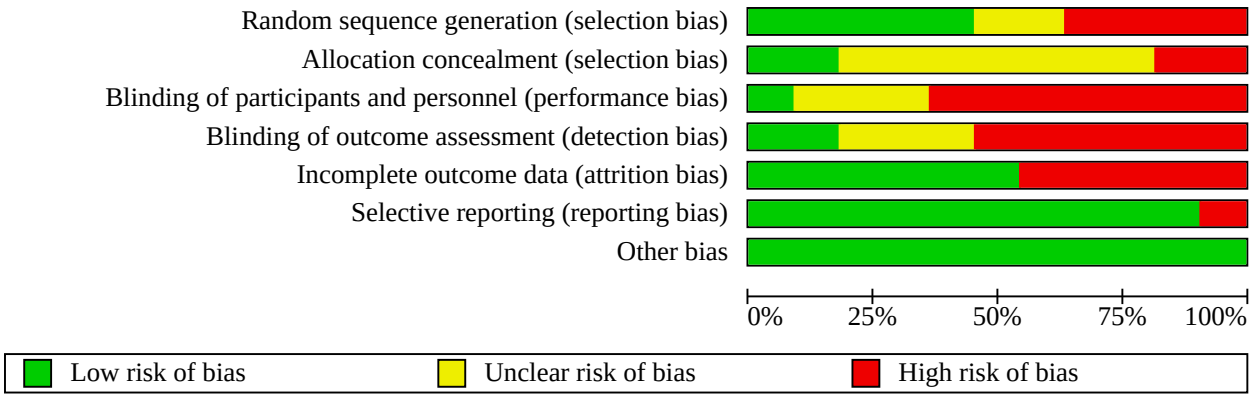

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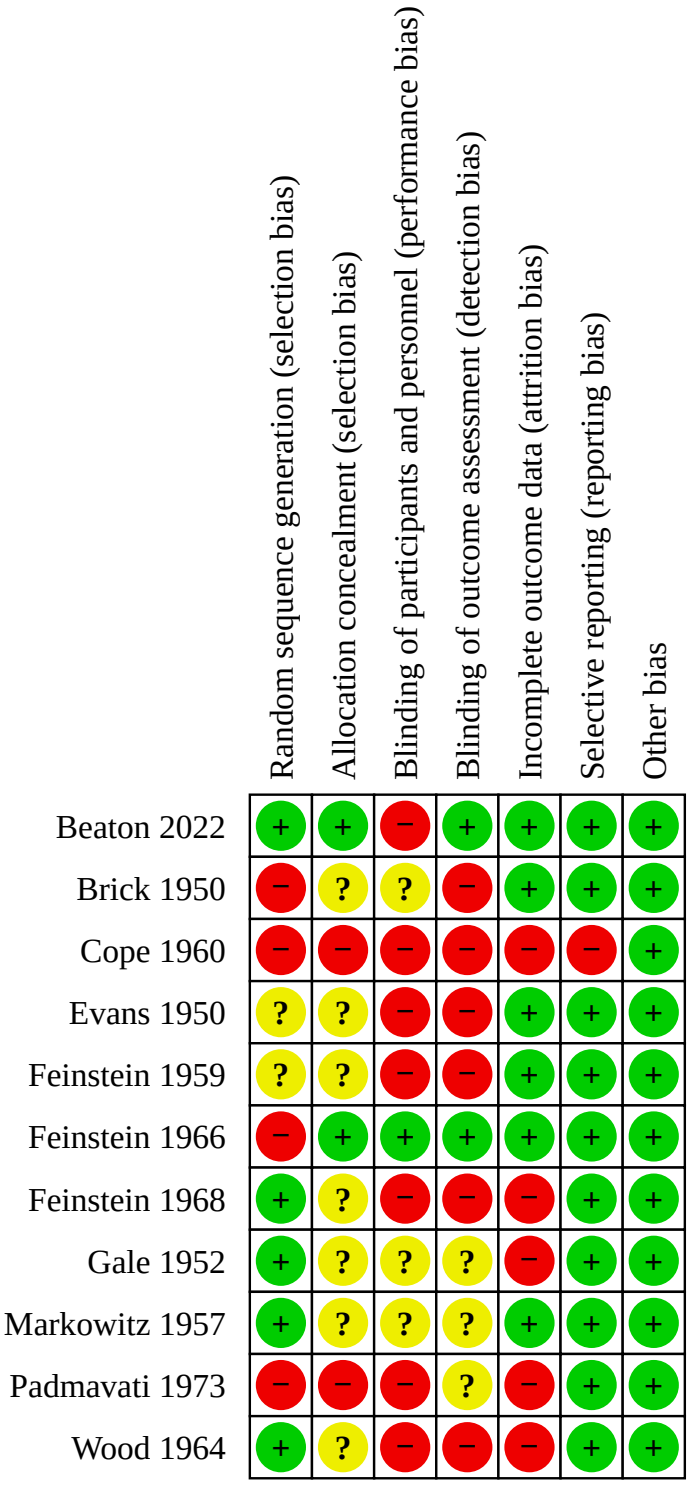
Figure 3. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.



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Random sequence generation (selection bias)

Padmavati 1973 allocated participants based on the days of the week, Brick 1950 alternated allocation to intervention and control, Feinstein 1966 assigned participants based on the date that they were admitted to the study in an alternating pattern, and Cope 1960 allocated participants to treatments based on the year of presentation, with those presenting in the first year of the study allocated to control, and those presenting in the second and third years allocated to Penicillin V and Penicillin G, respectively. Thus, we considered these four trials to be quasi-randomised and at high risk of bias. Evans 1950 and Feinstein 1959 did not state how randomisation occurred and so we considered these to be at unclear risk of bias. We considered the remaining five studies to be at low risk of bias for the following reasons. Beaton 2022 used a robust method of randomisation, using a web-based electronic data capture system managed by an independent statistician, with adequate allocation concealment. Gale 1952 paired participants in respect of sex, age and date of last attack of rheumatic fever, then allocated one from each pair to the penicillin and control groups by tossing a coin. Wood 1964 and Feinstein 1968 allocated participants to a series of consecutively numbered, sealed envelopes by use of statistical tables of random numbers. Markowitz 1957 allocated based on the last digit of their history numbers.

Allocation concealment (selection bias)

Allocation in Feinstein 1966 was concealed with penicillin and placebo coded and placed in cardboard boxes, with only the supply officer having knowledge of the contents. Beaton 2022 had an automated system for allocation concealment. Thus, we rated these studies at low risk of bias. Concealment of allocation was not clear in seven of the other included studies (Brick 1950; Evans 1950; Feinstein 1959; Feinstein 1968; Gale 1952; Markowitz 1957; Wood 1964). Prior to allocation, Wood 1964 and Feinstein 1968 stratified participants based on age, cardiac status and duration of freedom from rheumatic disease, and therefore it is unclear if allocation was truly concealed. There appeared to be no allocation concealment in two studies (Cope 1960; Padmavati 1973), so we rated these at high risk of bias for this domain.

Blinding (performance bias and detection bias)

The only study with adequate blinding of participants and outcome assessors was Feinstein 1966, as only the supply officer of the placebo and penicillin in this trial knew the identity of the contents of treatment boxes. Gale 1952 blinded participants by giving placebo tablets in bottles identical to those used in the treatment arm; however, it is not mentioned if other trial staff or outcome assessors were blinded. It was unclear whether Brick 1950 or Markowitz 1957 applied appropriate blinding procedures. Seven studies inadequately blinded participants, as they compared an oral medication versus no placebo (Cope 1960; Evans 1950) or compared intramuscular treatments versus oral (Beaton 2022; Feinstein 1959; Feinstein 1968; Padmavati 1973; Wood 1964). It does not appear that outcome assessors were blinded in any of these studies except for Beaton 2022, as outcome assessors were often those administering treatments. Beaton 2022 adequately blinded outcome assessors, including assessment of serial echocardiograms, despite not being able to blind participants due to the nature of the intervention. Performance bias and detection bias therefore cannot be excluded due to lack of blinding of participants, key trial staff and outcome assessors. This is significant where outcomes are subjective, however most of the primary and secondary outcomes assessed were objective. Beaton 2022 was not downgraded for risk of bias concerning the outcome of progression of RHD as this was based on echocardiographic images interpreted by blinded assessors. Nevertheless, the diagnosis of recurrence of rheumatic fever or progression of RHD may be biased if strict diagnostic criteria are not applied, and outcome assessors are not blinded.

Incomplete outcome data (attrition bias)

We considered the risk of attrition bias to be low in six studies (Beaton 2022; Brick 1950; Evans 1950; Feinstein 1959; Feinstein 1966; Markowitz 1957), due to low and comparable attrition in the intervention and control groups. We judged five randomised trials at high risk of attrition bias (Cope 1960; Feinstein 1968; Gale 1952; Padmavati 1973; Wood 1964). In Cope 1960, participants were admitted and discharged from the trial at different times during the trial period, meaning the groups were different with regards to outcome data collected, creating a significant attrition difference. There was significant loss to follow-up in Gale 1952 and the authors acknowledge that this disproportionally affected the placebo group. In Feinstein 1968, there was evidence of a difference in prophylaxis adherence between the two groups and > 5% loss to follow-up. Attrition was not consistent between groups in Wood 1964, with significant loss to follow-up in this trial. Padmavati 1973 was considered at high risk due to a high proportion of loss to follow-up. There was also incomplete outcome data where participants were non-adherent to their assigned intervention. Adherence to treatment was not consistently reported in all studies.

Selective reporting (reporting bias)

To determine the risk of reporting bias, we compared intended outcomes outlined in the Methods section with those reported in the Results section of published reports, as original trial protocols were not available in all cases except Beaton 2022. Ten studies reported all intended outcomes and are at low risk of reporting bias (Beaton 2022; Brick 1950; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Padmavati 1973; Wood 1964). Cope 1960 primarily provided a narrative of outcomes of participants included in the study. Participants were admitted and discharged at different times throughout the trial period; thus, outcomes appeared incomplete and we rated this study at high risk of bias.

Other potential sources of bias

No other potential sources of bias were identified in the 11 included studies, so we rated all at low risk of other bias (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Padmavati 1973; Wood 1964).

Effects of interventions

Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis

See Summary of findings 1.

None of the seven studies included in this comparison (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973) assessed our secondary outcomes of obstetric complications (maternal events and foetal events), treatment adherence or acceptability to participants.

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Primary outcomes

Recurrence of rheumatic fever

Six included studies provided data on recurrence of rheumatic fever (Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973). In combined analysis, moderate-certainty evidence suggests that antibiotics (intramuscular or oral) probably reduce the risk of recurrence of rheumatic fever by 61% (risk ratio (RR) 0.39, 95% confidence interval (CI) 0.22 to 0.69; I^2 = 0%, P = 0.83, tau^2 = 0; 6 studies, 1721 participants; Analysis 1.1), compared to no antibiotic prophylaxis. The number needed to treat to benefit (NNTB) is 33 (95% CI 25 to 100). There was a paucity of information on the specific oral antibiotic used in each study.

Subgroup analysis

Rheumatic fever with carditis

There was no evidence that risk of recurrence is higher in individuals with previous attacks of carditis (Chi^2 = 0.14, P = 0.70, I^2 = 0%; Analysis 1.1).

Duration of treatment

As antibiotics were given for the duration of each included study, using study follow-up duration as a proxy for antibiotic duration with meta-regression, we found that study duration was not associated with probability of recurrence of rheumatic fever (coefficient 0.01, 95% CI -0.02 to 0.05; P = 0.43; 6 studies, 1721 participants; Figure 4; Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973).

Figure 4. Meta-regression of risk of recurrence of rheumatic fever from Analysis 1.1 versus study follow-up (as proxy for antibiotic duration).

```
Random-effects meta-regression                  Number of obs   =         6
Method: REML                                    Residual heterogeneity:
                                                           tau2 = 4.6e-07
                                                         I2 (%) =      0.00
                                                             H2 =      1.00
                                                  R-squared (%) =      0.00
                                                Wald chi2(1)    =      0.62
                                                Prob > chi2     =    0.4296

------------------------------------------------------------------------------
    _meta_es | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    FollowUp |   .0130862    .016566     0.79   0.430    -.0193827     .045555
       _cons |  -1.592005   .8281863    -1.92   0.055     -3.21522    .0312105
------------------------------------------------------------------------------
Test of residual homogeneity: Q_res = chi2(4) =  1.40    Prob > Q_res = 0.8447
```

Class of antibiotic

In subgroup analysis, there was no evidence of a difference between the oral antibiotics subgroup and the intramuscular antibiotics subgroup (Chi^2 = 0.36, df = 1 (P = 0.55), I^2 = 0%; Analysis 1.1). There is insufficient evidence that oral benzylpenicillin (Chi^2 0.01, P = 0.92, I^2 = 0%) or oral phenoxymethylpenicillin (Chi^2 = 1.55, P = 0.21, I^2 = 35%) reduce recurrence of rheumatic fever compared with other included antibiotics (analysis not shown).

Sensitivity analysis

Oral benzylpenicillin

Benzylpenicillin (penicillin G) is now less frequently used via the oral route. Nevertheless, in sensitivity analysis, we found that oral antibiotics (excluding oral benzylpenicillin), compared with no antibiotic prophylaxis, still likely reduce recurrence of rheumatic fever by a similar effect magnitude to our overall results (RR 0.41, 95% CI 0.23 to 0.76; I^2 = 0%, P = 0.70, tau^2 = 0; 4 studies, 566 participants; Brick 1950; Evans 1950; Gale 1952; Padmavati 1973; analysis not shown).

Diagnosis of rheumatic fever

Cope 1960 provides data using both mJC and clinical judgement. In sensitivity analysis, including recurrences diagnosed using both definitions, there is evidence that oral antibiotics, compared with no antibiotic prophylaxis, probably reduce the risk of recurrence of rheumatic fever by 57% (RR 0.43, 95% CI 0.26 to 0.72; I^2 0%, P = 0.91, tau^2 = 0; 6 studies, 1721 participants; Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973). Furthermore, the diagnosis of recurrence of rheumatic fever using the mJC versus using clinical consensus alone did not make a difference to whether antibiotics, compared with no antibiotic prophylaxis, were effective (Chi^2 = 0.06, I^2 0%, P = 0.81; analysis not shown).

Peer-reviewed publications

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All included studies were peer-reviewed and no studies were considered to be completely at low risk of bias. With the removal of quasi-randomised trials, the confidence intervals became so wide that our primary outcome was no longer significant (RR 0.13, 0.02 to 1.03, I^2 = 0%, P = 0.85, tau^2 = 0; 2 studies, 373 participants; Evans 1950; Gale 1952; analysis not shown).

Progression of rheumatic heart disease

Latent RHD

The only study to include specifically latent RHD was Beaton 2022. This study demonstrated that intramuscular benzathine benzylpenicillin, compared with no antibiotic prophylaxis, likely reduces the risk of recurrence of RHD progression considerably, by approximately 91% (RR 0.09, 95% CI 0.03 to 0.29; 1 study, 818 participants; moderate-certainty evidence; Analysis 1.2; Beaton 2022). The NNTB is 14 (95% CI 10 to 20). This outcome was not downgraded for risk of bias due to blinding participants as the assessors interrogating the echocardiographic images were blinded.

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to the use of modified intention-to-treat (mITT) analysis (RR 0.09, 95% CI 0.03 to 0.29; 1 study, 799 participants; Beaton 2022; analysis not shown).

Late-stage RHD

One study collected data on the risk of progression of RHD in a population with 79.9% pre-existing chronic valvular heart disease that was considered late-stage RHD. From these data it was unclear whether antibiotics, compared with no antibiotic prophylaxis, reduce the recurrence of rheumatic fever (RR 1.50, 95% CI 0.66 to 3.40; 1 study, 994 participants; Analysis 1.3; Padmavati 1973).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Sensitivity analysis was not feasible due to there being only one included study.

Cardiac complications

Cardiac complications in the form of carditis were noted by one included study (Padmavati 1973). There may be no evidence from this study that antibiotics, compared with no antibiotic prophylaxis, reduce the risk of carditis (RR 0.94, 95% CI 0.69 to 1.27; 1 study, 994 participants; Analysis 1.4; Padmavati 1973).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Sensitivity analysis was not feasible due to there being only one included study.

Secondary outcomes

Mortality

One study reports low-certainty mortality data in latent RHD that antibiotics, compared with no antibiotic prophylaxis, may not affect all-cause mortality (RR 0.33, 95% CI 0.03 to 3.19; 1 study, 818 participants; Analysis 1.5; Beaton 2022).

Similarly, one study reports low-certainty mortality data in late-stage RHD showing that antibiotics, compared with no antibiotic prophylaxis, may not affect all-cause mortality (RR 1.23, 95% CI 0.78 to 1.94; 1 study, 994 participants; low-certainty evidence; Analysis 1.6; Padmavati 1973).

Subgroup analysis

Subgroup analysis was not feasible due to insufficient relevant data.

Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to use of mITT analysis (RR 0.09, 95% CI 0.03 to 0.29; 1 study, 799 participants; Beaton 2022; analysis not shown). All trials were peer-reviewed.

Adverse events

Anaphylaxis

Evidence is scarce, but antibiotics may not affect the risk of anaphylaxis, compared to no antibiotics, when used in an attempt to reduce recurrence of rheumatic fever (Peto OR 7.39, 95% CI 0.15 to 372; 1 study, 818 participants; low-certainty evidence; Analysis 1.7; Beaton 2022).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to use of mITT analysis (Peto OR 7.41, 95% CI 0.15 to 373; 1 study, 799 participants; Beaton 2022; analysis not shown).

Sciatic nerve injury

Evidence is scarce, but antibiotics may not affect the risk of sciatic nerve injury, compared to no antibiotic prophylaxis, when intramuscular benzathine benzylpenicillin is used in an attempt to reduce recurrence of rheumatic fever (Peto OR 7.39, 95% CI 0.15 to 372; 1 study, 818 participants; low-certainty evidence; Analysis 1.8; Beaton 2022).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

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Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to use of mITT analysis (Peto OR 7.41, 95% CI 0.15 to 373; 1 study, 799 participants; Beaton 2022; analysis not shown).

Delayed hypersensitivity or allergic reaction

There was evidence that using antibiotics to reduce recurrence of rheumatic fever is likely associated with more delayed hypersensitivity or allergic reactions compared with no antibiotic prophylaxis (RR 137, 95% CI 8.51 to 2205; 2 studies, 894 participants; moderate-certainty evidence; Analysis 1.9; Beaton 2022; Brick 1950). Brick 1950 did not report any events.

Subgroup analysis

Subgroup analysis was not feasible due to insufficient relevant data.

Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to use of mITT analysis (RR 137, 95% CI 8.53 to 2210; 2 studies, 875 participants; Beaton 2022; Brick 1950; analysis not shown).

Quasi-randomised trials and contemporary trials

Exclusion of one quasi-randomised trial, leaving one contemporary trial, resulted in the same result of increased delayed hypersensitivity or allergic reactions compared with no antibiotic prophylaxis, as Brick 1950 contributed no events (RR 137, 95% CI 8.51 to 22052; 1 study, 818 participants; moderate-certainty evidence; Beaton 2022; analysis not shown).

Local reactions to injection

Data from Beaton 2022 show that intramuscular benzathine benzylpenicillin likely leads to a significantly greater risk of having a local reaction defined as "redness, bruising or bleeding" (RR 29, 95% CI 1.74 to 485; 1 study, 818 participants; moderate-certainty evidence; Analysis 1.10; Beaton 2022).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Per-protocol analysis

Use of data from per-protocol analysis is very similar to use of mITT analysis (RR 29, 95% CI 1.74 to 486; 1 study, 799 participants; Beaton 2022; analysis not shown).

Comparison 2 - Intramuscular penicillin versus oral antibiotic

None of the four studies included in this comparison (Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964) assessed our secondary outcomes of cardiac complications or obstetric complications (maternal events and foetal events).

Primary outcomes

Recurrence of rheumatic fever

Meta-analysis of two studies reporting count data and two studies reporting rate data consistently showed that prophylactic intramuscular benzathine benzylpenicillin likely reduces recurrence of rheumatic fever substantially when compared to oral antibiotics (0.1% versus 1%, respectively) (count data: RR 0.07, 95% CI 0.02 to 0.26, I^2 = 0%, P = 0.68, tau^2 = 0; 2 studies, 395 participants; moderate-certainty evidence; Analysis 2.1; Markowitz 1957; Wood 1964; and rate data: rate ratio 0.12, 95% CI 0.04 to 0.33, I^2 = 0%, P = 0.80, tau^2 = 0; 2 studies, 769 participants; moderate-certainty evidence; Analysis 2.2; Feinstein 1959; Feinstein 1968). Count data NNTB 8 (95% CI 4 to 50).

Subgroup analysis

Rheumatic fever with carditis

Little information was available on whether recurrences featured carditis. Wood 1964 defined recurrence of rheumatic fever if there was evidence of carditis, and there was no evidence that antibiotics affected these individuals differently from individuals for whom carditis was not specified to be a feature of recurrence (Chi^2 = 0.16, P = 0.69, I^2 = 0%; Analysis 2.1).

Class of antibiotic

Wood 1964 used a combination of oral penicillin and sulfadiazine (with the results of each group reported independently, see Stollerman 1955) and Markowitz 1957 used oral penicillin. Comparing the subset of results from Wood 1964 using sulfadiazine with Markowitz 1957 showed no difference between oral penicillin and sulfadiazine subgroups (Chi^2 = 0.24, P = 0.63, I^2 = 0%; analysis not shown).

Sensitivity analysis

Oral benzylpenicillin

With the removal of studies using oral benzylpenicillin, from count data, there was still evidence that intramuscular benzathine benzylpenicillin is superior to oral antibiotics (RR 0.07, 95% CI 0.02 to 0.27; 1 study, 289 participants; Analysis 2.1; Wood 1964; analysis not shown).

Progression of rheumatic heart disease

From the evidence available, it is unclear whether intramuscular benzathine benzylpenicillin is superior to oral antibiotics at reducing the risk of progression of RHD (RR 0.28, 95% CI 0.03 to 2.22; 1 study, 256 participants; Analysis 2.3; Wood 1964). Participants had clinical RHD, not latent RHD.

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Sensitivity analysis was not feasible due to there being only one included study.

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Secondary outcomes

Mortality

It is unclear whether intramuscular benzathine benzylpenicillin is superior to oral antibiotics when used as a prophylactic agent at reducing the risk of all-cause mortality from rheumatic fever (Peto OR 0.22, 95% CI 0.01 to 4.12; 1 study, 431 participants; very low-certainty evidence; Analysis 2.4; Wood 1964).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Sensitivity analysis was not feasible due to there being only one included study.

Treatment adherence

From two studies in outpatient clinic settings, when intramuscular benzathine benzylpenicillin was used instead of oral antibiotics, participants were more likely to adhere to prophylaxis for rheumatic fever recurrence (RR 0.12, 95% CI 0.06 to 0.22; 2 studies, 577 participants; Analysis 2.5; Markowitz 1957; Wood 1964).

Subgroup analysis

Class of antibiotic

There was no evidence for a difference in adherence when comparing oral sulphasalazine or oral penicillin versus intramuscular penicillin ($Chi^2 = 0.41$, P = 0.52, $I^2 = 0\%$; analysis not shown).

Sensitivity analysis

Oral benzylpenicillin

With the removal of one study using oral benzylpenicillin, there was still evidence that intramuscular benzathine benzylpenicillin is associated with better adherence compared with oral antibiotics (RR 0.12, 95% CI 0.06 to 0.24; 1 study, 471 participants; Wood 1964; analysis not shown).

Adverse events: local reactions to injection

In one study, local reactions were noted in 13 participants who received intramuscular benzathine benzylpenicillin (10.9%) (Feinstein 1968).

Subgroup analysis

Subgroup analysis was not feasible due to there being only one included study.

Sensitivity analysis

Sensitivity analysis was not feasible due to there being only one included study.

Acceptability to participants: refusal of intramuscular injections

Feinstein 1968 randomised participants to receive intramuscular or oral antibiotics and reported the number of participants who refused either intervention. Of the 119 participants assigned to intramuscular antibiotic administration, four initially refused and were transferred to an oral regime, and 13 were transferred after painful local reactions (Feinstein 1968).

Subgroup analysis

Subgroup analysis was not feasible due to insufficient relevant data.

Sensitivity analysis

Sensitivity analysis was not feasible due to insufficient relevant data.

DISCUSSION

Summary of main results

See Summary of findings 1; Summary of findings 2.

This systematic review and meta-analysis investigated the prophylactic effect of antibiotics overall versus no antibiotics and intramuscular antibiotics versus oral antibiotics in individuals with previous rheumatic fever or rheumatic heart disease (RHD). There is moderate-certainty evidence that antibiotics as a whole are probably associated with a reduction of almost two-thirds in the risk of recurrence of rheumatic fever (Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973). Moreover, there is moderate-certainty evidence that intramuscular benzathine benzylpenicillin is probably substantially superior to oral antibiotics (by approximately 10 times) (Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964). Intramuscular benzathine benzylpenicillin likely reduces the risk of progression of RHD in individuals with latent RHD (Beaton 2022). It must be noted that the evidence on latent RHD is from only one study each with a two-year duration (Beaton 2022), and this has implications for the generalisability of the data. Further, this must be balanced against evidence that intramuscular benzathine benzylpenicillin, when compared with no antibiotics, likely causes a significantly greater number of local reactions to injection (Beaton 2022) and antibiotics are probably associated with more delayed hypersensitivity or allergic reactions (Beaton 2022; Brick 1950). Antibiotics may not affect all-cause mortality in latent RHD (Beaton 2022). Evidence is scarce, but antibiotics compared with no antibiotics may not affect the risk of anaphylaxis or sciatic nerve injury (Beaton 2022). It is unclear if intramuscular benzathine benzylpenicillin is superior to oral antibiotics for all-cause mortality (Wood 1964). When comparing intramuscular antibiotics to oral antibiotics, there were no data available on progression of latent RHD or adverse events including anaphylaxis, sciatic nerve injury, delayed hypersensitivity/allergic reactions and local reactions to injection. It is important to interpret these findings in the context of major limitations, including the following: the vast majority of included studies were conducted more than 50 years ago, many before contemporary echocardiographic studies; most were of low quality with methodology often at high risk of bias; outdated treatments were used; there was only one study in latent RHD; and there are concerns regarding generalisability to low socioeconomic regions.

Overall completeness and applicability of evidence

This review constitutes the most up-to-date and complete appraisal of evidence for the use of antibiotics as prophylactic agents for rheumatic fever recurrence and RHD. The search was comprehensive and was performed in the CENTRAL, MEDLINE, Embase and CPCI-S databases and in clinical trials registries

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(ClinicalTrials.gov, WHO ICTRP, ISRCTN). Given that most trials on the topic were performed several decades ago, some of the older articles were difficult to obtain, although there were no citations identified that we were unable to gain sufficient information from. The available evidence exceeds what has been established previously and addresses our primary questions, but there remain issues with the applicability of the evidence to contemporary challenges. Much of the evidence identified is historic, from trials published over 50 years ago with methodological and randomisation issues. As a result, there remains a significant gap in contemporary high-quality evidence. Since the publication of most of the studies (nine) in the 1950s-1960s (Brick 1950; Cope 1960; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Wood 1964), the use of echocardiography has revolutionised the diagnosis of sub-clinical disease. The modern role of echocardiography in the detection and prevention of progression of RHD is not captured in many of the included historic trials, again emphasising the need for high-quality contemporary evidence. As a result, the study of latent RHD is a new paradigm of study that requires further work. Only two included studies were conducted in low-income countries where the current challenges are greatest. Globally, the prevalence of RHD is highest in south Asia and sub-Saharan Africa, with rates of > 1000 cases per 100,000 population and the participants in these trials only partially represent the demographic most likely to be impacted by the intervention (Watkins 2017). Four of the studies were completed in designated care settings for children with rheumatic fever, three in the UK (Cope 1960; Evans 1950; Gale 1952), and one in New York, USA (Feinstein 1966). There was a paucity of data available to answer most of our a priori defined subgroup and sensitivity analyses. The timing and duration of antibiotic prophylaxis was inconsistent across these trials, and at times as short as one month (Evans 1950). Interestingly, as many of the included studies were historical, two described using oral benzylpenicillin (penicillin G) (Cope 1960; Feinstein 1966). This formulation is rarely used orally in modern practice due to poor bioavailability (Yip 2022), but sensitivity analysis demonstrated that removal of these studies from our results had little impact on our effect estimates.

Nonetheless, our outcomes of interest covered the majority of the data available in the included studies. Few studies assessed the role of penicillin prophylaxis in the progression of RHD or associated cardiac complications. Only Beaton 2022 analysed the efficacy of penicillin in preventing latent RHD progression by echocardiographic assessment. Other studies did not comment on RHD progression (Cope 1960; Evans 1950; Gale 1952), or only discussed cardiac disease progression in cases of acute rheumatic fever recurrence rather than in the cohort as a whole (Feinstein 1966). Additionally, cardiac complications analysed were often limited to rheumatic fever associated carditis (Cope 1960; Feinstein 1966; Padmavati 1973) or worsening valvular heart disease (Beaton 2022). There is limited available evidence for review authors to comment on other cardiac complications relating to RHD, namely arrhythmias or endocarditis.

There were only three included studies that met the criteria for inclusion in comparison two. Four RCTs were included (Feinstein 1959; Feinstein 1968; Markowitz 1957; Wood 1964), all of which analysed the recurrence of streptococcal throat infections and acute rheumatic fever in paediatric populations who were allocated to receive intramuscular or oral antibiotics.

A key limitation of this review is the generalisability of findings to the population who bear the greatest burden of disease, with only two out of 11 included studies conducted in low-income countries. Similarly, there was variety in the settings where the included studies were conducted. Several studies were conducted in an inpatient or convalescent home setting (Cope 1960; Evans 1950), but the place of care of such patients has evolved since the publication of these historical trials, which may further limit the generalisability of findings to current times. Furthermore, only one study investigated latent RHD (Beaton 2022). Although this study was recent and of high quality, the lack of other studies investigating this specific population limits the generalisability of our findings.

All but one (Beaton 2022) of the studies in this review were historical in nature, conducted between 1950 and 1973, when the methodology of trials was not as rigorously conducted or reported as it is today. Subsequently, we judged all included studies to be at high risk of bias in at least one domain, particularly with regard to blinding and follow-up (attrition bias). The criteria used for the diagnosis of rheumatic fever and relapses were often not described in the methodology of these studies and may have varied. Furthermore, the duration of antibiotics was not explicitly stated in four studies (Brick 1950; Evans 1950; Feinstein 1966; Gale 1952) and therefore we used the duration of the follow-up period as a proxy for duration of antibiotic therapy. The most commonly used penicillins were benzylpenicillin and phenoxymethylpenicillin; however, two studies did not state which type of penicillin was used (Brick 1950; Gale 1952). Finally, there was heterogeneity in the reporting of adverse outcomes, and it is possible that not all adverse outcomes were investigated for or reported. In addition to the benefits of intramuscular benzathine benzylpenicillin laid out in this review, intramuscular benzathine benzylpenicillin may contribute towards regression of valve lesions (Torres 2021). Nonetheless, prior to the use of intramuscular benzathine benzylpenicillin, careful thought and discussion is required. There are variable formulations and practice around injection delivery, and risks appear lower and patient experience better with pre-formulated liquid, use of local anaesthetic and the ventro-gluteal route for administration (Ralph 2021).

There are several additional non-randomised studies on this topic, which were outside the scope of this review, and many of which are detailed in this review in the Characteristics of excluded studies table. The findings of many of these studies are summarised in the review by Ambari 2024.

Certainty of the evidence

Comparison one - antibiotics versus no antibiotics

We considered all included studies to be at high risk of bias for at least one domain (Beaton 2022; Brick 1950; Cope 1960; Evans 1950; Feinstein 1966; Gale 1952; Padmavati 1973); we considered four studies to be quasi-randomised (Brick 1950; Cope 1960; Feinstein 1966; Padmavati 1973), and thus at high risk for random sequence generation. We considered two studies to be at high risk for allocation concealment (Cope 1960; Padmavati 1973), three to have an unclear risk of bias (Brick 1950; Evans 1950; Gale 1952) and two to have a low risk of bias (Beaton 2022; Feinstein 1966). We assessed four studies to be at high risk of bias for blinding of participants and personnel (Beaton 2022; Cope 1960; Evans 1950; Padmavati 1973) and three studies to be at high risk of bias for outcome assessment

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(Brick 1950; Cope 1960; Evans 1950). We judged two studies to be at unclear risk of bias for blinding of participants and personnel (Brick 1950; Gale 1952). We considered one study at unclear risk of bias (Gale 1952) and one study at low risk of bias (Beaton 2022) for outcome assessment. Regarding attrition bias, we assessed three studies to be at high risk (Cope 1960; Gale 1952; Padmavati 1973) and four at low risk (Beaton 2022; Brick 1950; Evans 1950; Feinstein 1966). We considered one study to be at high risk of selective reporting bias (Cope 1960). We did not identify any other bias.

We integrated risk of bias into our GRADE findings. We downgraded the outcome 'Recurrence of rheumatic fever' by one level for risk of bias as all but one study was unblinded and for other concerns over risk of bias as detailed above. For the outcome 'Progression of RHD (latent RHD)', we downgraded the certainty of the evidence by one level for imprecision as only one study contributed to the results. Whilst Beaton 2022 did not employ blinding throughout their study, it is worth noting that many of the outcomes used were well-described and objective. For the outcome 'Mortality (late-stage RHD)', we downgraded the certainty of the evidence by one level for risk of bias and one level for imprecision. We downgraded the certainty of the evidence for the outcome 'Anaphylaxis' and 'Sciatic nerve injury' by two levels for imprecision, as the results are based on one study with a very low event rate. We downgraded the evidence for the outcomes 'Delayed hypersensitivity or allergic reaction' and 'Local reactions to injection' for imprecision as the results were based on one study.

Comparison two - intramuscular penicillin versus oral antibiotics

There were four randomised included studies in this comparison, and one was considered at unclear risk of bias for random sequence generation due to insufficient information (Feinstein 1959). We considered all four studies at unclear risk of bias for allocation concealment. We considered three studies to be at high risk of bias for blinding of participants and personnel, and outcome assessment (Feinstein 1959; Feinstein 1968; Wood 1964), and one at unclear risk (Markowitz 1957). Regarding attrition bias, we assessed two studies to be at high risk (Feinstein 1968; Wood 1964) and two at low risk (Feinstein 1959; Markowitz 1957). We did not identify any other bias.

We downgraded the outcome'Recurrence of rheumatic fever' by one level for risk of bias. We downgraded the outcome 'Mortality' by one level for risk of bias, one level for imprecision and one level for indirectness. We downgraded for indirectness because RHD is a progressive disease and the effect on mortality is likely to be delayed, even beyond the endpoint of the trial.

Potential biases in the review process

This review was conducted in accordance with the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011a; Higgins 2011b; Higgins 2022; Higgins 2023a; Schünemann 2011; Schünemann 2023), where steps were taken to reduce the risk of bias in the review process, wherever possible. However, due to the length of time since some of the studies have been published and due to the fact that some of them may have been published in journals that are no longer in print, it is possible some studies may have been missed. We tried to overcome this with an extensive search. Again, due to the age of the majority of included studies and the changes in the delivery of health care during this time, some aspects of the methodology of these studies were not always clear. We have endeavoured to accurately interpret these reports, but there is a possibility of misinterpretation. This review was designed with wide inclusion criteria. Whilst having wide inclusion criteria improves generalisability, it risks missing specific circumstances where antibiotic prophylaxis may be most beneficial. We have sought to ameliorate this factor through the use of subgroup analysis and by splitting the severity of RHD when considering progression, thus looking at the bigger picture whilst not losing sight of specific circumstances.

It was not possible to analyse outcomes at different time points as separate comparisons to avoid a unit of analysis error. Given that we were not able to pool more than 10 studies for any of our outcomes, we were unable to assess for publication bias.

Agreements and disagreements with other studies or reviews

The key findings of this review are consistent with those of the last Cochrane review in this area from over two decades ago (Manyemba 2002). To our knowledge, that review constitutes the only previous review in this area and it included nine studies. However, the authors of this review were surprised to find four new RCTs that were published prior to 2002 but were not included within the previous review on this topic (Brick 1950; Cope 1960; Evans 1950; Gale 1952). This may have been because they only searched the Controlled Trials Register, MEDLINE and Embase; however, Brick 1950 was excluded from the previous review when it was found, as it was considered to be non-randomised. We carried out a more extensive search, which is detailed in our search strategy. We have also included quasi-randomised trials, including Brick 1950, which alternated allocation to intervention and control. Manyemba and Mayosi concluded in favour of penicillin compared to no treatment for the prevention of progression to RHD (Manyemba 2002). This analysis was based on 1301 patients from three studies. Despite not having addressed this point in particular in this review, Manyemba 2002 looked at two or three versus four-weekly regimens to understand which might be more effective, with data coming from single studies performed in the 1990s. We are not aware of any further RCTs looking at this question.

A rapid review commissioned by the Norwegian Institute of Public Health has previously sought to address the use of secondary prophylaxis in RHD or rheumatic fever, but with a focus on penicillin (Fønhus 2020). The review included four RCTs and two cohort trials, and did not include Cope 1960, Feinstein 1959, Feinstein 1968 and Wood 1964, even though they all used a penicillin as one of their interventions and were conducted before 2020. No information is provided on excluded studies or why these studies were not included. Fønhus 2020 found similar results to this review in that they reported that: 1) prophylactic penicillin probably reduces rheumatic fever recurrences, with moderate certainty, and 2) there is uncertain evidence regarding the effects of prophylactic penicillin on mortality. However, Fønhus 2020 had different findings; firstly, they report that there was no evidence on progression of RHD, in contrast to this review that found two studies reporting this outcome, including one before 2020 (Padmavati 1973). Secondly, Fønhus 2020 reports uncertain (very low-certainty) evidence of adverse events with prophylactic use of antibiotics in RHD, but this review includes new data from Beaton 2022 that show that antibiotics likely increase the risk of delayed hypersensitivity or allergic reactions. The GOAL trial, Beaton 2022, is the principal

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addition to the literature since both reviews, being the only RCT to assess intramuscular penicillin prophylaxis compared to placebo control in latent RHD. The positive findings of this study add significant weight to arguments in favour of penicillin prophylaxis for reduction of progression of latent RHD. The trial duration was only two years, but longer-term data may be published in the next couple of years. However, the relatively poor quality of much of the previously available evidence limits the strength of conclusions.

A recent systematic review aimed at summarising preventative interventions for RHD including antibiotics since 2000 found evidence from observational data consistent with substantial decreases in rheumatic fever recurrence with antibiotic prophylaxis (Shimanda 2024).

The findings of the current review remain in favour of the assertion that intramuscular penicillin is likely superior to oral penicillin and align with the conclusions of Manyemba and Mayosi (Manyemba 2002) that well-designed RCTs comparing the two interventions are required to confirm the superiority of either route of administration in preventing RHD and associated cardiac complications. This review found that antibiotics, overall, administered via oral and intramuscular routes, likely reduce the risk of recurrence of rheumatic fever compared to control. Moreover, there is now evidence from one study that intramuscular benzathine benzylpenicillin likely reduces progression of RHD in individuals with latent RHD when compared to oral antibiotics. Manyemba and Mayosi also concluded in favour of intramuscular penicillin administration over oral, following their analysis of four studies with 1098 participants. The results were consistent across all four studies. The available evidence published following Manyemba and Mayosi's review focuses on associated elements of intramuscular penicillin therapy, rather than its increased efficacy compared to oral, namely allergy, side effects and adherence. Anaphylaxis remains a rare but highly concerning adverse effect of intramuscular penicillin, as detailed by Çiftel et al (Çiftel 2015). Reassuringly, Kaya et al demonstrated extremely low rates of true penicillin allergy in their evaluation of 535 children receiving intramuscular penicillin (Kaya 2014). Similarly, improving adherence to regular penicillin therapy is a key area for research, with reviews analysing factors influencing or predicting good compliance with prophylaxis (Dixit 2023; Kevat 2017). Further reviews have assessed the benefits of antibiotic prophylaxis being effectively integrated into a country's health system, rather than as standalone initiatives (Abrams 2020). Rémond et al have discussed the difficulties in creating and implementing effective programmes in this regard (Rémond 2016). These reviews demonstrate that the focus and direction of research is in optimising the delivery and maintenance of intramuscular penicillin prophylaxis with further work being done on two-weekly versus four-weekly injections (de Dassel 2019). Timing of antibiotic delivery was not an area we focused on during this review but is a topic demanding further research to determine the optimum time frame for disease prevention and control.

AUTHORS' CONCLUSIONS

Implications for practice

This review provides evidence from randomised controlled trials (RCTs) and quasi-RCTs that antibiotic prophylaxis compared to no antibiotics likely reduces the risk of recurrence of rheumatic fever and that intramuscular benzathine benzylpenicillin is probably superior to oral antibiotics (recurrence of rheumatic fever is likely lower with intramuscular benzathine benzylpenicillin than oral antibiotics). Intramuscular benzathine benzylpenicillin likely reduces the risk of progression of latent rheumatic heart disease (RHD). Antibiotics are probably associated with more delayed hypersensitivity or allergic reactions compared with no antibiotics. Evidence is scarce, but antibiotics compared with no antibiotics may not affect the risk of anaphylaxis or sciatic nerve injury. Antibiotics may not affect mortality in late-stage RHD. There is little evidence available to comment on the effect of intramuscular penicillin over oral antibiotics on mortality and no evidence for progression of latent RHD and adverse events.

It is important to note that the evidence that intramuscular benzathine benzylpenicillin versus no antibiotics likely reduces the risk of progression of latent RHD is based on a single trial (Beaton 2022), in one sample of African school children with latent mild RHD detected by echocardiographic screening. It is also important to note that Beaton 2022 did not show that antibiotic prophylaxis had a significant impact on echocardiographic regression from latent (Stage A-B) RHD at two years. This underlines the need for ongoing research to understand who most benefits from prophylaxis.

Whilst intramuscular benzathine benzylpenicillin appears substantially superior to oral antibiotics (approximately 10 times more effective), this should be balanced against the risks of delayed hypersensitivity or allergic reactions, and local reactions to the injection, although, at present, the risk of significant adverse reactions, such as anaphylaxis, appears to be low. Further, whilst it appears that antibiotic prophylaxis reduces rheumatic fever recurrences, there are few data on other endpoints. People with early or mild RHD likely have the greatest capacity to benefit from prophylaxis. Individuals with more severe RHD may have less capacity to benefit.

Implications for research

This review has demonstrated the lack of research in this area over the past several decades. With ever-increasing globalisation, it is likely that the proportion of individuals living with the sequelae of RHD will rise globally. Generalisability has probably become the biggest limitation in the evidence surrounding this area at present, and future work must endeavour to include populations who bear the greatest burden of this disease. Furthermore, most of the included studies identified in this review were historical in nature (nine trials from the 1950s-1960s; Brick 1950; Cope 1960; Evans 1950; Feinstein 1959; Feinstein 1966; Feinstein 1968; Gale 1952; Markowitz 1957; Wood 1964), when the methodology of trials was not built as rigorously as it is today and, as such, many of the included studies have been classified at high risk of bias, particularly with regard to blinding and follow-up (attrition bias). Trials of oral antibiotics (e.g. amoxicillin, azithromycin, etc.) versus intramuscular penicillin in monitored settings seem to be of interest for this clinical scenario. Indeed, the GOALIE trial is underway (NCT05693545); this is a randomised trial comparing twice-daily oral penicillin prophylaxis versus monthly intramuscular penicillin in preventing latent RHD progression in children 5 to 17 years. Reporting the follow-up of pregnant women with secondary prophylaxis should also be encouraged. Of greater importance is the performance of trials aimed at increasing adherence to secondary rheumatic fever prophylaxis. Furthermore, these trials should include strategies to improve adherence and compliance with oral antibiotics. It would currently be unethical to

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perform trials that withhold antibiotics in patients with RHD at risk of rheumatic fever recurrence. In accordance with antimicrobial stewardship and in order to safeguard the efficacy of antibiotics for future generations, further work can be done to better understand the implications of long-term antibiotic prophylaxis for rheumatic fever and RHD for antibiotic resistance.

Future research should investigate endpoints for which we could not find any data (comparison one - antibiotics versus no antibiotics: obstetric complications, poor treatment adherence, diarrhoea and/or vomiting; comparison two - intramuscular penicillin versus oral antibiotics: cardiac complications, obstetric complications, anaphylaxis and diarrhoea and/or vomiting). In addition, future research should further investigate endpoints for which there was insufficient evidence to draw conclusive judgements (comparison one - antibiotics versus no antibiotics: progression of RHD (late-stage), cardiac complications, all-cause mortality, anaphylaxis; comparison two - intramuscular penicillin versus oral antibiotics: progression of RHD, all-cause mortality, acceptability - refusal of intramuscular injections). There were few data available to enable subgroup analyses. Further work should provide information to enable assessment of data based on the severity of RHD, the presence of carditis, age stratification, the duration of treatment and information relevant to pregnancy.

Whilst there is now better evidence for antibiotics as a whole, particularly including intramuscular benzathine benzylpenicillin, oral penicillin and oral sulfadiazine, there is currently little evidence to compare these and other classes of antibiotics. Moreover, the vast majority of the data presented in this review concerns penicillin, and to enable production of further guidelines tailored to providing antibiotic prophylaxis for RHD, it is necessary to consider other antibiotic alternatives; for example, for those who are allergic to penicillins. Future trials on other antibiotics apart from penicillin for secondary prevention of RHD are required.

Future work must take account of the fact that most of the populations to whom prophylaxis is relevant are young and may be of child-bearing age. Finally, latent RHD is poorly researched. In this review we identified one trial including individuals with latent RHD. Further work in this area should include individuals with latent RHD who, based on the findings of this review, appear to have the most to gain from antibiotic prophylaxis.

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Editorial and peer reviewer contributions

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- Sign-off Editor (final editorial decision): Gerald Gartlehner, Cochrane Austria, Danube University Krems;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Joanne Duffield, Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer reviewer comments and supported the editorial team): Lisa Wydrzynski, Central Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision): Joselyn Rwebembera, Division of Adult Cardiology, Uganda Heart Institute, Kampala, Uganda (clinical/content review); Clair Mills (clinical/content review); Bruno Ramos Nascimento, MD, MSc, PhD, FACC, FESC, Faculdade de Medicina, Universidade Federal de Minas Gerais, Brazil (clinical/content review); Associate Professor Rachel Webb, University of Auckland (clinical/content review); Jessica D'Urbano (consumer review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Steve McDonald, Cochrane Australia (search review). The following two peer reviewers commented on an earlier draft of this review: Rosangela Stadnick Lauth de Almeida Torres PhD, Professor at the Department of Medicine at Universidade Positivo; Coordinator of the National Reference Laboratory for Research and Identification of Streptococci, PR Brazil (clinical/content review); and Dr Rosemary Wyber, Telethon Kids Institute and Australian National University (clinical/content review).

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* Indicates the major publication for the study

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CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Beaton 2022

Study characteristics	
Methods	**Study design:** phase III, parallel-group, partially blinded (outcomes adjudicators blinded; patients, data collectors, local practitioners not blinded), pragmatic, single-site, randomised controlled trial **Total study duration:** 2 years **Number of study centres:** single centre **Study location:** Uganda **Study setting:** primary and secondary schools in Gulu and surrounding districts **Date of study:** 26 June 2018 to October 2020
Participants	**Number (n) of participants:** 818 **Mean sex:** 44% male overall (intervention = 43% male, control = 46% male) **Mean age:** not reported (range 5 to 17 years) **Inclusion criteria:** children aged 5 to 17, new diagnosis of latent RHD based on echo **Type of housing:** 19.9 = permanent, 79.2 = semi-permanent **Number (n) of persons living in the household:** 7.9 **Number (n) of persons < 15 years living in the household:** 3.75 **Type of school:** 87.2% day school, 12.7% boarding school **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Average duration of maternal education:** 5 years **Income (WAMI) index:** 0.3 **Sore throat reported in < 4 weeks:** 17.72% **Skin infection reported < 4 weeks:** 6.35% **At least 1 first-degree family relative with a previous diagnosis of RF:** 1.30% **At least 1 first-degree family relative with a previous diagnosis of RHD:** 2.30% **Previous acute RF:** 0% **History of RHD:** 0% **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** IM penicillin G benzathine 1.2 million IU (> 30 kg) to 600,000 IU (< 30 kg) monthly. BPG administration occurred at 4-week intervals for 26 x 4-week periods (24 months)

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Beaton 2022 *(Continued)*

	Control: no prophylaxis
Outcomes	**Primary outcome(s)** 1. Progression of RHD (latent RHD) at 24 months follow-up. The WHF 2012 criteria were used for the diagnosis of this using echocardiography. **Secondary outcome(s)** 1. Echocardiographic regression of latent RHD at 24 months 2. Adverse events
Notes	**Funding for studies:** Supported by the Thrasher Research Fund, Gift of Life International, Children's National Hospital Foundation (Zachary Blumenfeld Fund and Race for Every Child (Team Jocelyn)), the Elias–Ginsburg Family, Wiley Rein, Philips Foundation, AT&T Foundation, Heart Healers International, the Karp Family Foundation, Huron Philanthropies and the Cincinnati Children's Hospital Heart Institute Research Core. **Notable conflicts of interest of study authors:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Yes	**Quote:** "The randomization scheme was designed by an independent statistician and embedded within the randomization module of REDCap, a Web-based electronic data capture system."
Allocation concealment (selection bias)	Yes	**Quote:** "Two designated members of the research staff enrolled participants after consent or assent was obtained, at which time the trial-group assignment was revealed automatically."
Blinding of participants and personnel (performance bias)	No	**Quote:** "Injections of penicillin G benzathine were administered by trial staff who were trained in best practices. Participants in the control group received no prophylaxis and no placebo". **Comment:** Patients and personnel likely both aware of what treatment they received.
Blinding of outcome assessment (detection bias)	Yes	**Quote:** "The echocardiograms obtained at enrollment and at 2 years were presented side by side to the panel, with random right or left display. The members of the panel, who were unaware of the trial-group assignments and the timing of the echocardiograms"
Incomplete outcome data (attrition bias)	Yes	**Comment:** A total of 799 participants (97.7%) completed the trial: 399 in the prophylaxis group and 400 in the control group. No significant attrition in either group.
Selective reporting (reporting bias)	Yes	**Quote:** "Since only 3% of the data from the primary outcome assessments were missing, we analyzed the observed data."
Other bias	Yes	No other forms of bias were found in the paper.

Brick 1950

Study characteristics	
Methods	**Study design:** quasi-randomised trial

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Brick 1950 *(Continued)*

	Total study duration: 2 years **Number of study centres:** single **Study location:** Canada **Study setting:** outpatient clinic **Date of study:** autumn 1946
Participants	**Number (n) of participants:** 76 **Sex:** not reported **Average age:** 11 years **Age range:** children **Inclusion criteria:** "rheumatic patients, being followed in the cardiac clinic at the Hospital for Sick Children", "rheumatic fever patients were attending the cardiac clinic." **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** not reported **History of RHD:** not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** oral penicillin, 50,000 units twice daily **Control:** no treatment No information on the length of study.
Outcomes	1. Throat cultures positive for B haemolytic streptococci at 2 years follow-up 2. Number of upper respiratory infections at 2 years follow-up 3. Recurrence of rheumatic fever at 2 years follow-up 4. Adverse reactions

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Brick 1950 *(Continued)*

	The tools and definitions used for this were not clearly defined apart from the fact that throat swabs were used for diagnosing streptococcal infections. Due to the mention of chorea, polyarthritis, etc., Jones criteria were likely used. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** not reported **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	No	**Quote:** "In the fall of 1946, approximately 80 rheumatic fever patients were attending the cardiac clinic. These were divided into two groups by placing patients alternately in control and in penicillin groups. Eventually there were 38 children in each group and these two groups were comparable in all respects."
Allocation concealment (selection bias)	Unclear	**Comment:** No information available on concealment
Blinding of participants and personnel (performance bias)	Unclear	**Comment:** Blinding of participants and personnel is not given in the paper and was unlikely due to the type of intervention, which was oral vs no drug.
Blinding of outcome assessment (detection bias)	No	**Comment:** Blinding of assessors is not mentioned due to the type of intervention and assessors likely were aware. The assessors were the same as those giving treatment.
Incomplete outcome data (attrition bias)	Yes	**Comment:** No loss to follow-up in both groups.
Selective reporting (reporting bias)	Yes	**Comment:** Data collection appears to be complete.
Other bias	Yes	**Comment:** No other sources of bias were identified in the paper.

Cope 1960

Study characteristics	
Methods	**Study design:** quasi-randomised trial **Total study duration:** not reported **Number of study centres:** single **Study location:** UK **Study setting:** inpatient - St Joseph's Children's Heart Hospital, Rainhill **Date of study:** not reported
Participants	**Number (n) of participants:** 117 **Sex:** not reported

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Cope 1960 *(Continued)*

	Average age: 38 **Age range:** not reported **Inclusion criteria:** The cases of rheumatic fever occurring during the 3 years were considered under 2 headings: group A, those who fit the Jones criteria (modified) as reported by the Royal College of Physicians (1957); that is, showing 2 major or 1 major and 2 minor manifestations; and group B, those who, while not fulfilling the criteria, were considered to have rheumatic fever by 3 physicians who each saw the case independently. **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** not reported **History of RHD:** not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** oral penicillin V and penicillin G, 200,000 units twice daily for 1 year **Control:** no treatment
Outcomes	1. Cases of rheumatic fever at 1-year follow-up 2. Cases of chorea at 1-year follow-up 3. Cases of carditis at 1-year follow-up 4. Recurrence of rheumatic fever at 1-year follow-up For diagnosis of rheumatic fever, Jones criteria 1957 were used with either 2 major or 1 major and 2 minor criteria or alternatively 3 physicians agreeing on the diagnosis of rheumatic fever after seeing them independently. Throat swabs were used and the anti-streptolysin titres were tested in a lab. No further details were given in this regard, such as units. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** not reported **Notable conflicts of interest of study authors:** not reported

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Cope 1960 *(Continued)*

Power calculations: not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	No	**Comment:** Allocation to treatment was based on year of presentation, with first year being control, second year Penicillin V and third year Penicillin G.
Allocation concealment (selection bias)	No	**Quote:** "Control Period.-During this phase there were three cases of group A type. One had been in hospital for some years, the others had been admitted 14 weeks and eight weeks previously. Penicillin V Period.-One child had rheumatic fever of group A type as shown by fever, joint pains, pallor, malaise, and the appearance of a pericardial rub. This child had been well and in St. Joseph's for the previous 15 months. Three children had rheumatic fever of group B type." **Comment:** There appears to have been no concealment of allocation. It is not clear if allocation was concealed as only one type of treatment was offered per year. According to the *Cochrane Handbook*, "Other participants may be deliberately directed to the 'appropriate' intervention, which can often be accomplished by delaying a participant's entry into the trial until the next appropriate allocation appears".
Blinding of participants and personnel (performance bias)	No	**Comment:** Not clear if participants or personnel were blinded and not mentioned in the paper, but highly unlikely as some did not have treatment (control) and some had treatment (oral groups).
Blinding of outcome assessment (detection bias)	No	**Comment:** It is not mentioned in the paper if any of the data collectors were blinded, but unlikely as they were the same as those giving treatment.
Incomplete outcome data (attrition bias)	No	**Comment:** Patients were admitted and discharged during the study periods, making the three groups different regarding outcome data and creating significant attrition differences.
Selective reporting (reporting bias)	No	**Comment:** Patients were admitted and discharged in the reporting periods, thus, outcomes are not complete.
Other bias	Yes	**Comment:** No other sources of bias are reported in the paper.

Evans 1950

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 23 months **Number of study centres:** single **Study location:** UK **Study setting:** "Children's Heart Home, Lancing", a convalescent home for children **Date of study:** 9 November 1946 to 30 September 1948
Participants	**Number (n) of participants:** 300

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Evans 1950 *(Continued)*

	Sex: not reported **Average age:** not reported **Age range:** 5 to 13 **Inclusion criteria:** children of 5 to 13 years of age; Jones criteria not used **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** not reported **History of RHD:** not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** 100,000 units of calcium penicillin in 1 oz of 5% glucose three-quarters of an hour before breakfast **Control:** no treatment
Outcomes	1. Incidence of haemolytic streptococcal carriers 2. Incidence of clinical streptococcal infection 3. The occurrence of rheumatic relapses or evidence of fresh rheumatic activity Monthly anti-streptolysin titre was taken through venepuncture and weekly nose and throat swabs for streptococcal infections. It appears the diagnosis of rheumatic fever was made only on the basis of these 2 tests. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** not reported **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

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Evans 1950 *(Continued)*

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear	**Quote:** "On admission each child was allocated by random sampling to penicillin or control group"
Allocation concealment (selection bias)	Unclear	**Comment:** Allocation concealment is not mentioned in the paper.
Blinding of participants and personnel (performance bias)	No	**Comment:** Blinding of participants and personnel is not given in the paper and was unlikely due to the type of intervention, which was oral vs no drug.
Blinding of outcome assessment (detection bias)	No	**Comment:** Blinding of assessors is not mentioned due to the type of intervention and assessors were likely aware. The assessors were the same as those giving treatment.
Incomplete outcome data (attrition bias)	Yes	**Comment:** Attrition levels appear to have been low in both groups.
Selective reporting (reporting bias)	Yes	**Comment:** Data collection appears to be complete.
Other bias	Yes	**Comment:** No other sources of bias were identified in the paper.

Feinstein 1959

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 3 years **Number of study centres:** single **Study location:** USA **Study setting:** clinic **Date of study:** 1955
Participants	**Number (n) of participants:** 391 **Sex:** not reported **Average age:** not reported **Age range:** children, not otherwise specified **Inclusion criteria:** children, unequivocal episodes of ARF by modified Jones criteria **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/ sanitation:** not reported

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Feinstein 1959 *(Continued)*

Assets: not reported

Maternal education: not reported

Income (WAMI) index: not reported

Average duration of maternal education: not reported

Sore throat reported in < 4 weeks: not reported

Skin infection reported < 4 weeks: not reported

At least 1 first-degree family relative with a previous diagnosis of RF: not reported

At least 1 first-degree family relative with a previous diagnosis of RHD: not reported

Previous acute RF: not reported

History of RHD: not reported

WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study: not reported

Other comorbidities such as NYHA class at baseline and end of study: not reported

Interventions	**Intervention:** IM benzathine penicillin G 1.2 million units every 4 weeks for 3 years **Control** 1. Oral sulfadiazine 1 g/day 2. Oral buffered potassium penicillin G 200,000 units/day
Outcomes	1. Occurrence of streptococcal infection 2. Recurrence of rheumatic fever 3. Fidelity of prophylaxis Throats swabs for streptococci and blood tests for anti-streptolysin titres were used. Diagnosis was made on the basis of the 1955 Jones Criteria. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** AHA, NYHA, Westchester Heart Association, the Sullivan County Heart Chapter, USPHS, Wyeth Laboratories **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear	**Comment:** No information available.
Allocation concealment (selection bias)	Unclear	**Comment:** As the method of randomisation is not given, it is also unclear if allocation was concealed
Blinding of participants and personnel (performance bias)	No	**Comment:** Blinding of participants and personnel is not mentioned and unlikely as IM vs oral medications.

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Feinstein 1959 *(Continued)*

Blinding of outcome assessment (detection bias)	No	**Comment:** Blinding of outcome assessment is not mentioned in the paper and unlikely as those who treated patients also collected data.
Incomplete outcome data (attrition bias)	Yes	**Comment:** Attrition seems to have been similar in all three groups.
Selective reporting (reporting bias)	Yes	**Comment:** Reporting appears to be complete.
Other bias	Yes	**Comment:** There is no other bias reported in this paper.

Feinstein 1966

Study characteristics	
Methods	**Study design:** quasi-randomised trial **Total study duration:** 28 months **Number of study centres:** single **Study location:** USA **Study setting:** outpatient clinic **Date of study:** 5 January 1962
Participants	**Number (n) of participants:** 161 **Sex:** not reported **Average age:** 17.5 **Age range:** children **Inclusion criteria:** (1) a past unequivocal episode of acute rheumatic fever, fulfilling the modified Jones diagnostic criteria; (2) no current clinical evidence of rheumatic heart disease; (3) aged 14 or older; and (4) freedom from an acute attack of rheumatic fever for at least 5 years if aged 14 or 15, and for at least 3 years if aged 16 or older. **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported

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Feinstein 1966 *(Continued)*

	At least 1 first-degree family relative with a previous diagnosis of RF: not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** 100% **History of RHD:** not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** oral potassium penicillin G in tablets of 200,000 units once daily **Control**: placebo
Outcomes	1. Occurrence of streptococcal infection 2. Fidelity of prophylaxis 3. Rheumatic fever recurrences Jones criteria 1955 were used for diagnosis of rheumatic recurrences. Throat swabs for streptococci and anti-streptolysin titre was taken every 2 months. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** Public Health Service research grant CD 00030 from the National Institutes of Health and Dr. Jonas's work was supported by a Rheumatic Fever Fellowship of the New York State Department of Health. **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	No	**Comment:** Assigned based on date admitted to the study in an alternating pattern . **Quote:** ''Within these two groups, the "E" or "O" prophylaxis preparation was then assigned alternatingly as each patient was admitted to the study.''
Allocation concealment (selection bias)	Yes	**Quote:** "The penicillin and the placebo were received, coded, and placed in small cardboard boxes for distribution by a supply officer of Irvington House, the only person who knew the identity of the contents. The boxes were marked "E" (for even) and "O" (for odd), and the same medication (penicillin or placebo) was consistently given the same label." **Comment:** The allocation concealment was hidden as when the patients were assigned their group it was not known which drug they would be given.
Blinding of participants and personnel (performance bias)	Yes	**Comment:** As per the previous comment, participants and personnel were unaware of allocation.
Blinding of outcome assessment (detection bias)	Yes	**Comment:** Outcome assessors were unaware of the allocation.

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Feinstein 1966 *(Continued)*

Incomplete outcome data (attrition bias)	Yes	**Comment:** Similar attrition was found in both groups.
Selective reporting (reporting bias)	Yes	**Comment:** Outcomes appear to be reported completely.
Other bias	Yes	**Comment:** There are no other sources of bias reported.

Feinstein 1968

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 4 years **Number of study centres:** single **Study location:** USA **Study setting:** tertiary hospital **Date of study:** November 1962
Participants	**Number (n) of participants:** 278 **Sex:** not reported **Average age:** 11 **Age range:** 5 to 16 **Inclusion criteria:** children, unequivocal episodes of ARF by modified Jones criteria **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/ sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** all had previous ARF > 3 years previously as per modified Jones criteria

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Feinstein 1968 *(Continued)*

	History of RHD: not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** IM penicillin G benzathine 1.2 million units once monthly for 4 years **Control:** oral penicillin G potassium, 400,000 units 3 times daily
Outcomes	1. Occurrence of streptococcal infection 2. Fidelity of prophylaxis 3. Rheumatic fever recurrences Every month, patients were seen and a history was taken, a throat examination was done and swab was taken for streptococcal infection. Every 2 months an anti-streptolysin titre was taken. Jones criteria 1955 was used to diagnose rheumatic fever. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** Public Health Service research grant CD 00030 from the National Institutes of Health and Dr Jonas's work was supported by a Rheumatic Fever Fellowship of the New York State Department of Health **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Yes	**Comment:** Used and quoted randomisation method from the study Wood 1964. **Quote:** "Three of these variables were selected as most important for planning the statistical randomisation: age, cardiac status and duration of freedom from rheumatic activity...The patient's status in relation to these three variables was used to place him in one of the eight groups shown in this table...For each of the eight different groups, A through H, a series of consecutively numbered envelopes was prepared. Each sealed envelope contrained the name of the one of the TWO (two in Feinstein 1968, but 3 drugs in Wood 1964 paper) drugs, allocated to the envelopes by use of statistical tables of random numbers. As each patient entered the study, his position in Table 1 was determined, then the corresponding envelope ws opened in sequence, indicating drug to be used. For example the fifth patient to be admitted to Group B received the drug named in envelope B-5."
Allocation concealment (selection bias)	Unclear	**Comment:** Not clear if allocation was concealed due to stratified randomisation.
Blinding of participants and personnel (performance bias)	No	**Comment:** Blinding is not mentioned and due to the comparison (i.e. injection vs tablets), it would have been difficult to do.
Blinding of outcome assessment (detection bias)	No	**Comment:** Blinding of assessors is not mentioned but unlikely as assessors were likely the same as those giving treatment, which was IV vs oral.

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Feinstein 1968 *(Continued)*

Incomplete outcome data (attrition bias)	No	**Comment:** There was significant difference in prophylaxis adherence between the two groups, and > 5% loss to follow-up.
Selective reporting (reporting bias)	Yes	**Comment:** Outcomes appear to be completely reported.
Other bias	Yes	**Comment:** No other sources of bias reported.

Gale 1952

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 8 months **Number of study centres:** single **Study location:** UK **Study setting:** outpatient clinic **Date of study:** 14 October 1950
Participants	**Number (n) of participants:** 73 **Sex:** penicillin group 34.1% male, control group 31.3% male **Average age:** 11 **Age range:** 6 to 15 **Inclusion criteria:** children with rheumatism and heart disease **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** not reported

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Gale 1952 *(Continued)*

	History of RHD: not reported **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** oral penicillin 200,000 units twice a day, length of course unspecified **Control:** placebo
Outcomes	1. Positive swab for group A strep 2. Recurrence/relapse of rheumatic fever Throat swabs were taken every 2 weeks. Diagnosis of rheumatic fever was made on the result of the swab. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** grant from the Royal College of Physicians **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Yes	**Comment:** tossing a coin, so low risk as per risk of bias guidance **Quote:** "They were paired as far as possible in respect of sex, age, and date of last attack, and one of each pair was allotted to the penicillin and one to the lactose group by tossing a coin. This was done before the parents were asked if they were willing to help in the experiment"
Allocation concealment (selection bias)	Unclear	**Comment:** Unclear if allocation was concealed.
Blinding of participants and personnel (performance bias)	Unclear	**Quote:** "The children in the lactose group were given their tablets in a bottle exactly like that used for penicillin tablets and were also given the glucose." **Comment:** It is not mentioned if assessors were blinded even though children were.
Blinding of outcome assessment (detection bias)	Unclear	**Comment:** It is not mentioned if outcome assessors were blinded.
Incomplete outcome data (attrition bias)	No	**Quote:** "When the parents were invited to take part they were not told to which group their child had been allocated, but unfortunately more of those allocated to the penicillin group accepted. Another difficulty in securing perfectly matched groups was that some of the children discharged from Winford lived too far from Bristol to attend the outpatient department regularly; and it so happened that there were more of these in the lactose than in the penicillin group." **Comment:** There was significant dropout in the trial. Allocation was more in one geographical area than the other, which was further, thus creating attrition. Attrition was far more in one group than the other.

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Gale 1952 *(Continued)*

Selective reporting (reporting bias)	Yes	**Comment:** The expected outcomes are all included, so despite the attrition, the outcomes available were reported completely. According to the *Cochrane Handbook*, "The study protocol is not available but it is clear that the published reports include all expected outcomes, including those that were prespecified (convincing text of this nature may be uncommon)." (Schünemann 2023)
Other bias	Yes	**Comment:** There were no other reported sources of bias.

Markowitz 1957

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 14 months **Number of study centres:** single **Study location:** USA **Study setting:** outpatient clinic **Date of study:** September/October 1954 to November 1955
Participants	**Number (n) of participants:** 114 **Mean sex:** 42.9% male in IM group, 54.0% male in oral group **Mean age:** 11.7 years (IM group), 11.5 years (oral group) **Inclusion criteria:** children, diagnosis of rheumatic fever or rheumatic heart disease by 'acceptable criteria' **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 years living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Average duration of maternal education:** not reported **Income (WAMI) index:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** 100%

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Markowitz 1957 *(Continued)*

	History of RHD: 28/56 in IM group and 22/50 in oral group **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention (N = 61) IM:** penicillin G benzathine 1,200,000 units, 4 times weekly, for the duration of the study (up to 14 months) **Control (N = 53):** oral penicillin G benzathine 200,000 units once a day before breakfast, for the duration of the study (up to 14 months) "Patients in both groups were instructed to return once a month. The patients were examined at each clinic visit, throat cultures were taken, and blood was drawn for ASO titer and sedimentation rate. Fluoroscopic examination and electrocardiograms were obtained whenever indicated"
Outcomes	1. Recurrence of rheumatic fever 2. Streptococcal infections without rheumatic sequale 3. Isolation of Group A streptococcus from throat 4. Adverse reactions NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** funded by the Heart Association of Maryland and the Benjamin and Minnie Landsberg Memorial Foundation **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Yes	**Quote:** "Two groups were established by allocating patients according to the last digit of their history numbers: odd numbers were assigned to the group receiving oral medication, even numbers to the group receiving the drug by the intramuscular route."
Allocation concealment (selection bias)	Unclear	**Comment:** Concealment unclear.
Blinding of participants and personnel (performance bias)	Unclear	**Comment:** Unclear if participants or investigator were blinded.
Blinding of outcome assessment (detection bias)	Unclear	**Comment:** Unclear if participants and personnel were blinded to assessment - no mention. Unclear if outcome assessors were blinded - no mention.
Incomplete outcome data (attrition bias)	Yes	**Comment:** 7% loss to follow-up, but similar across intervention and control (control 3/53 and intervention 5/61).
Selective reporting (reporting bias)	Yes	**Comment:** Outcome data appear to be complete but no protocol. Outcomes reported in methods are consistent with results.
Other bias	Yes	**Quote:** "This study was aided in part by the Heart Association of Maryland and the Benjamin and Minnie Landsberg Memorial Foundation."

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Padmavati 1973

Study characteristics	
Methods	**Study design:** quasi-randomised trial **Total study duration:** 5 years **Number of study centres:** single **Study location:** India **Study setting:** hospital **Date of study:** 1966
Participants	**Number (n) of participants:** 994 **Sex:** penicillin group 34.1% male, control group 31.3% male **Average age:** not reported **Age range:** a) up to 5 years, b) 5 to 9 years, c) 10 to 14 years, d) 15 to 19 years **Inclusion criteria:** any patient on suspicion of rheumatic fever and/or rheumatic carditis **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** public **Water/sanitation:** not reported **Assets:** income groups (socioeconomic status using currency at the time) - Rs 100 and below, Rs 101 to 250, Rs 251 to 500, Rs 501 to 750, Rs 751 to 1000, Rs 1000 and above **Maternal education:** not reported **Income (WAMI) index:** used currency Rs instead, with poorest income groups, so Rs 250 or below being the majority of the participants in the study **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** no timeline given but Table V on symptoms of first attack of RF - sore throat reported as the first attack in 42% (197/327) of patients in the antibiotic group and 42% (191/250) of patients in control group **Skin infection reported < 4 weeks:** no timeline given - no patient has erythema marginatum on first attack; 4/327 patients (antibiotics group) and 3/250 (control) have subcutaneous nodules. **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** 577 **History of RHD:** 100% **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported

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Padmavati 1973 *(Continued)*

Interventions	**Intervention:** oral benzathine penicillin once a month, unspecified duration **Control:** penicillin and other antibiotics as and when needed for throat infection and vitamin B injections once a month
Outcomes	1. Streptococcal infections 2. Rheumatic fever recurrence 3. All-cause mortality All patients underwent a thorough clinical examination including ECGs, chest X-ray, routine haemogram and ESR. Throat and nasal swabs were taken for culture and blood collected for ASO and AH titrations once a month. ASO titrations are carried out by the log dilution method of Robinson, Crawford and Roholt. AH titrations are carried out by Quinn's method. Modified Jones criteria 1965 was used to diagnose rheumatic fever. NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** not reported **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	No	**Comment:** The study's method for assigning patients to either the prophylaxis or control group based on the day of the week does not adhere to standard random sequence generation practices, leading to potential predictability in allocation
Allocation concealment (selection bias)	No	**Comment:** Given the allocation of patients based on predetermined days without any mention of measures to conceal this allocation process, there is a significant risk that the assignment could have been anticipated, thereby compromising the study's integrity.
Blinding of participants and personnel (performance bias)	No	**Comment:** The absence of any described measures for blinding both the study participants and personnel to the intervention groups raises concerns about performance bias, as perceptions and behaviours could have been influenced by this knowledge.
Blinding of outcome assessment (detection bias)	Unclear	**Comment:** Without information on whether outcome assessors were blinded to the intervention groups, the potential for detection bias cannot be ruled out, as assessors' knowledge could affect outcome interpretation.
Incomplete outcome data (attrition bias)	No	**Comment:** The study reports a 25.8% dropout rate and significant irregular attendance, raising concerns about attrition bias. This high rate of incomplete outcome data could skew the results, undermining the reliability of the findings.
Selective reporting (reporting bias)	Yes	**Comment:** The study does not provide information on how it addressed the possibility of selective reporting, leaving the risk of this bias unclear. Without transparency on whether all pre-specified outcomes were reported, there is potential for reporting bias. However, outcomes reported in the methods were consistent with the results.
Other bias	Yes	**Comment:** There were no other reported sources of bias.

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Wood 1964

Study characteristics	
Methods	**Study design:** randomised controlled trial **Total study duration:** 6 years **Number of study centres:** single **Study location:** USA **Study setting:** clinic **Date of study:** 1 May 1954
Participants	**Number (n) of participants:** 431 **Sex:** 62.9% male **Average age:** 11.4 **Age range:** 5 to 18 **Inclusion criteria:** children and adolescents 5 to 18 years old, unequivocal episode **Type of housing:** not reported **Number (n) of persons living in the household:** not reported **Number (n) of persons < 15 living in the household:** not reported **Type of school:** not reported **Water/sanitation:** not reported **Assets:** not reported **Maternal education:** not reported **Income (WAMI) index:** not reported **Average duration of maternal education:** not reported **Sore throat reported in < 4 weeks:** not reported **Skin infection reported < 4 weeks:** not reported **At least 1 first-degree family relative with a previous diagnosis of RF:** not reported **At least 1 first-degree family relative with a previous diagnosis of RHD:** not reported **Previous acute RF:** 100% **History of RHD:** 40.60% **WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study:** not reported **Other comorbidities such as NYHA class at baseline and end of study:** not reported
Interventions	**Intervention:** IM benzathine penicillin G 1,200,000 units every 4 weeks, planned once per month for 6 years, mean duration in study was 3.6 years **Control:** oral buffered potassium penicillin G 200,000 units/day or oral sulfadiazine 1 g/day

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Wood 1964 *(Continued)*

Outcomes	1. Occurrence of streptococcal infection 2. Recurrence of rheumatic fever 3. Progression of rheumatic heart disease Modified Jones criteria 1955 were used to diagnose rheumatic fever. At each monthly clinic visit, careful interval histories were taken. A complete physical examination was done every 2 months and more often when indicated. Throat cultures were taken at each clinic visit. When beta haemolytic streptococci were isolated on throat culture, serologic grouping was carried out to determine whether or not the organism belonged to Lancefield Group A. Blood was taken from each patient every 2 months. The sera were routinely tested for antistreptolysin O (ASO) titre and then preserved in the frozen state. When indicated, blood was taken every month for several months rather than every 2 months. When bacteriologic or serologic suspicion of streptococcal infection arose, the sera of several consecutive months were thawed and tested simultaneously for antistreptolysin O, anti-hyaluronidase and antistreptokinase titres. This procedure was called a "serial run." NB: it is not specified whether these outcomes were primary or secondary
Notes	**Funding for studies:** NIHR grant H-1595, AHA, NYHA, Westchester Heart Association, Wyeth Laboratories, Sullivan County Heart Chapter and the Genesee County Heart Chapter **Notable conflicts of interest of study authors:** not reported **Power calculations:** not reported

Item	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Yes	**Quote:** "Three of these variables were selected as most important for planning the statistical randomisation: age, cardiac status and duration of freedom from rheumatic activity...The patient's status in relation to these three variables was used to place him in one of the eight groups shown in this table...For each of the eight different groups, A through H, a series of consecutively numbered envelopes was prepared. Each sealed envelope contrained the name of the one of the TWO (two in Feinstein 1968, but 3 drugs in Wood 1964 paper) drugs, allocated to the envelopes by use of statistical tables of random numbers. As each patient entered the study, his position in Table 1 was determined, then the corresponding envelope was opened in sequence, indicating drug to be used. For example the fifth patient to be admitted to Group B received the drug named in envelope B-5."
Allocation concealment (selection bias)	Unclear	**Comment:** It is not clear if allocation was known in advance due to the way randomisation was done, through putting patients in groups rather than true randomisation; stratification may affect allocation concealment.
Blinding of participants and personnel (performance bias)	No	**Comment:** The regimes were all different and compared IM and oral treatments. The oral treatments were also given at different times, so participants and personnel likely knew which group they were in. **Quote:** "The prophylactic regimens employed were: [1] benzathine penicillin G, 1,200,000 units IM every 4 weeks, [2] buffered potassium penicillin G, 200,000 units/day orally in a single dose, 1/2 hour before breakfast, and [3] sulfadiazine, 1 g/day orally in a single dose."
Blinding of outcome assessment (detection bias)	No	**Quote:** "When an unduly long interval elapsed between clinic visits, the clinic staff attempted to maintain supply of prophylaxis either by mailing additional pills or by making a home visit to give the injection when necessary."

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Wood 1964 *(Continued)*

		Comment: It appears the data collectors were also giving the treatment. This suggests that outcome assessment was not blinded.
Incomplete outcome data (attrition bias)	No	**Comment:** Attrition was not consistent between groups. There was significant dropout from the study. There was also significant movement of patients between arms. **Quote:** "These consisted of 80 patients (18 per cent of the total sample) of whom 20 were on benzathine penicillin, 33 on oral penicillin and 27 on sulfadiazine".
Selective reporting (reporting bias)	Yes	**Comment:** The outcomes appear to be completed in those who did not drop out. According to the *Cochrane Handbook*, "The study protocol is not available but it is clear that the published reports include all expected outcomes, including those that were pre-specified (convincing text of this nature may be uncommon)." (Schünemann 2023)
Other bias	Yes	**Comment:** There were no other reported risks of bias.

AH: antihyaluronidase
AHA: American Heart Association
ARF: acute rheumatic fever
ASO: antistreptolysin O titre
AH: antihyaluronidase
AT&T: American Telephone and Telegraph Company
BPG: benzathine penicillin G
ECG: electrocardiogram
ESR: erythrocyte sedimentation rate
IM: intramuscular
N/n: number
NIHR: National Institute for Health and Care Research
NYHA: New York Heart Association
Penicillin G: potassium, sodium injection
RF: rheumatic fever
RHD: rheumatic heart disease
Rs: Indian Rupee currency
vs: versus
WAMI: Water/sanitation, Assets, Maternal education, and Income
WHF: World Heart Federation

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Belay 2022	Ineligible intervention (no control)
Breese 1958	Ineligible population
Bywaters 1958	Ineligible study design (non-randomised study)
Chamovitz 1954	Ineligible population; no relevant outcomes of interest
Frank 1965	Ineligible study design (non-randomised study)
Hand 2019	Ineligible intervention (no control)
Hassell 1974	Ineligible study design

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Study	Reason for exclusion
Herdy 1996	Ineligible comparison (IM benzyl penicillin versus IV antibiotic then oral antibiotic)
Kassem 1992	Ineligible intervention (no control)
Kassem 1996	Ineligible comparison (comparing timing of intervention)
Kohn 1950	Ineligible study design (non-randomised study)
Lalchandani 2016	Ineligible intervention (no control)
Lue 1996	Ineligible comparison (comparing timing of intervention)
Roberts 1953	Ineligible intervention (no control)
Schreier 1958	Ineligible intervention (no control)
Spinetto 2011	Ineligible study design (non-randomised study)
Sprenger 2016	Ineligible study design
Stollerman 1955	Ineligible study design (non-randomised study)
Thomas 1988	Ineligible study design (non-randomised study)
Tompkins 1972	Ineligible study design (non-randomised study)

IM: intramuscular
IV: intravenous

Characteristics of ongoing studies *[ordered by study ID]*

NCT05693545

Study name	GOALIE: intramuscular vs enteral penicillin prophylaxis to prevent progression of latent RHD trial
Starting date	1 December 2023
Contact information	ndate.fall@cchmc.org
Methods	**Study design:** randomised controlled trial (phase III, parallel-group, partially blinded (outcomes adjudicators blinded; patients, data collectors, local practitioners not blinded), pragmatic, single-site, randomised, controlled, non-inferiority trial) **Total study duration:** 2 years (estimated completion 30 September 2027) **Number of study centres:** single **Study location:** Uganda **Study setting:** schools in Lira and surrounding districts in Northern Uganda **Date of study:** 1 September 2023
Participants	**Number (n) of participants:** 1004 (estimated) **Mean sex:** not reported

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NCT05693545 *(Continued)*

Mean age: not reported

Inclusion criteria

- Children aged 5 to 17 years
- Has a new diagnosis of latent RHD detected through primary or secondary school echocardiographic screening
- Has agreed to participate in the study via the study's informed consent/assent process

"Operational Definition of Latent RHD Borderline RHD or Mild Definite RHD (to include no more than mild regurgitation at the mitral or aortic valve, normal mean mitral and aortic valve gradients, normal bi-ventricular function) according to the 2012 WHF consensus criteria"

Type of housing: not reported

Number (n) of persons living in the household: not reported

Number (n) of persons < 15 years living in the household: not reported

Type of school: not reported

Water/sanitation: not reported

Assets: not reported

Maternal education: not reported

Average duration of maternal education: not reported

Income (WAMI) index: not reported

Sore throat reported in < 4 weeks: not reported

Skin infection reported < 4 weeks: not reported

At least 1 first-degree family relative with a previous diagnosis of RF: not reported

At least 1 first-degree family relative with a previous diagnosis of RHD: not reported

Previous acute RF: not reported

History of RHD: not reported

WHF (World Heart Federation) criteria RHD diagnosis data echocardiographic severity at baseline and end of study: not reported

Other comorbidities such as NYHA class at baseline and end of study: not reported

Interventions	**Intervention:** oral phenoxymethyl penicillin (Pen V) prophylaxis 250 mg twice daily **Control:** intramuscular benzathine benzylpenicillin G (BPG) prophylaxis (600,000 IU for children < 30 kg, 1.2 million IU for children ≥ 30 kg), every 28 days
Outcomes	**Primary outcome(s)** 1. Progression of echocardiographic features of latent RHD to borderline to definite, or definite to mild or definite to moderate/severe **Secondary outcome(s)** 1. Compare the economic equivalence and cost-effectiveness of oral penicillin compared to IM penicillin 2. Compare patient-reported outcomes (treatment acceptance, treatment satisfaction, and health-related quality of life)

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NCT05693545 *(Continued)*

Notes	**Funding for studies:** National Institutes of Health (NHLBI 5R33HL166441-03). Sponsor - Children's Hospital Medical Center, Cincinnati **Notable conflicts of interest of study authors:** none

IM: intramuscular
NYHA: New York Heart Association
RF: rheumatic fever
RHD: rheumatic heart disease
vs: versus
WAMI: Water/sanitation, Assets, Maternal education, and Income
WHF: World Heart Federation

DATA AND ANALYSES

Comparison 1. Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Recurrence of rheumatic fever	6	1721	Risk Ratio (M-H, Random, 95% CI)	0.39 [0.22, 0.69]
1.1.1 Oral antibiotics	5	727	Risk Ratio (M-H, Random, 95% CI)	0.32 [0.13, 0.79]
1.1.2 Intramuscular antibiotics	1	994	Risk Ratio (M-H, Random, 95% CI)	0.45 [0.22, 0.92]
1.2 Progression of rheumatic heart disease (latent RHD)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.3 Progression of rheumatic heart disease (late-stage RHD)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.4 Cardiac complications	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.5 All-cause mortality (latent RHD)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.6 All-cause mortality (late-stage RHD)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
1.7 Adverse events: anaphylaxis	1		Peto Odds Ratio (Peto, Fixed, 95% CI)	Totals not selected
1.8 Adverse events: sciatic nerve injury	1		Peto Odds Ratio (Peto, Fixed, 95% CI)	Totals not selected
1.9 Adverse events: delayed hypersensitivity or allergic reaction	2	894	Risk Ratio (M-H, Random, 95% CI)	137.00 [8.51, 2205.07]

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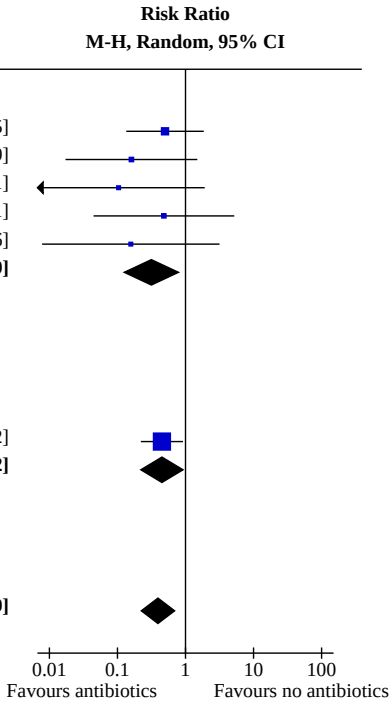
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.10 Adverse events: local reactions to injection	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

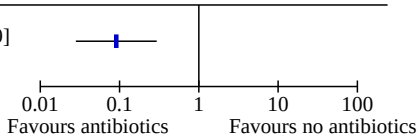
Analysis 1.1. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 1: Recurrence of rheumatic fever

Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Weight	Risk Ratio M-H, Random, 95% CI
1.1.1 Oral antibiotics						
Brick 1950	3	38	6	38	18.4%	0.50 [0.13 , 1.85]
Cope 1960	1	79	3	38	6.4%	0.16 [0.02 , 1.49]
Evans 1950	0	155	4	145	3.7%	0.10 [0.01 , 1.91]
Feinstein 1966	1	82	2	79	5.6%	0.48 [0.04 , 5.21]
Gale 1952	0	41	2	32	3.5%	0.16 [0.01 , 3.16]
Subtotal (95% CI)		**395**		**332**	**37.7%**	**0.32 [0.13 , 0.79]**
Total events:	5		17			
Heterogeneity: Tau² = 0.00; Chi² = 1.76, df = 4 (P = 0.78); I² = 0%						
Test for overall effect: Z = 2.47 (P = 0.01)						
1.1.2 Intramuscular antibiotics						
Padmavati 1973	11	523	22	471	62.3%	0.45 [0.22 , 0.92]
Subtotal (95% CI)		**523**		**471**	**62.3%**	**0.45 [0.22 , 0.92]**
Total events:	11		22			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.19 (P = 0.03)						
Total (95% CI)		**918**		**803**	**100.0%**	**0.39 [0.22 , 0.69]**
Total events:	16		39			
Heterogeneity: Tau² = 0.00; Chi² = 2.11, df = 5 (P = 0.83); I² = 0%						
Test for overall effect: Z = 3.25 (P = 0.001)						
Test for subgroup differences: Chi² = 0.36, df = 1 (P = 0.55), I² = 0%						



Analysis 1.2. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 2: Progression of rheumatic heart disease (latent RHD)

Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Risk Ratio M-H, Random, 95% CI
Beaton 2022	3	409	33	409	0.09 [0.03 , 0.29]



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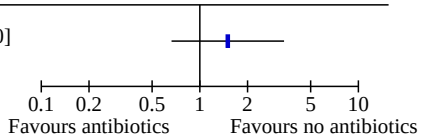
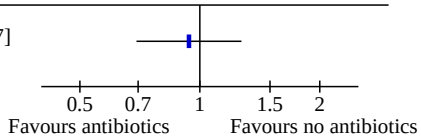
Analysis 1.3. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 3: Progression of rheumatic heart disease (late-stage RHD)

	Antibiotics		No antibiotics		Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	M-H, Random, 95% CI	M-H, Random, 95% CI
Padmavati 1973	15	523	9	471	1.50 [0.66 , 3.40]	



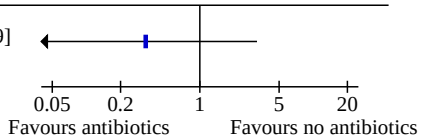
Analysis 1.4. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 4: Cardiac complications

	Antibiotics		No antibiotics		Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	M-H, Random, 95% CI	M-H, Random, 95% CI
Padmavati 1973	73	523	70	471	0.94 [0.69 , 1.27]	



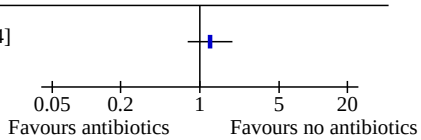
Analysis 1.5. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 5: All-cause mortality (latent RHD)

	Antibiotics		No antibiotics		Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	M-H, Random, 95% CI	M-H, Random, 95% CI
Beaton 2022	1	409	3	409	0.33 [0.03 , 3.19]	



Analysis 1.6. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 6: All-cause mortality (late-stage RHD)

	Antibiotics		No antibiotics		Risk Ratio	Risk Ratio
Study or Subgroup	Events	Total	Events	Total	M-H, Random, 95% CI	M-H, Random, 95% CI
Padmavati 1973	41	523	30	471	1.23 [0.78 , 1.94]	



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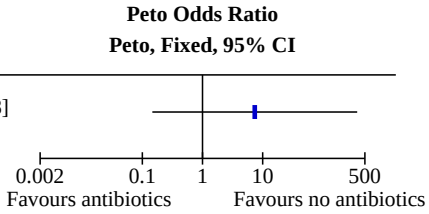
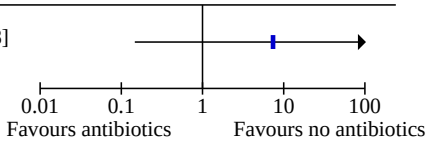
Analysis 1.7. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 7: Adverse events: anaphylaxis

Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Peto Odds Ratio Peto, Fixed, 95% CI
Beaton 2022	1	409	0	409	7.39 [0.15 , 372.38]



Analysis 1.8. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 8: Adverse events: sciatic nerve injury

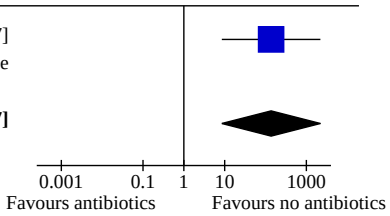
Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Peto Odds Ratio Peto, Fixed, 95% CI
Beaton 2022	1	409	0	409	7.39 [0.15 , 372.38]



Analysis 1.9. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 9: Adverse events: delayed hypersensitivity or allergic reaction

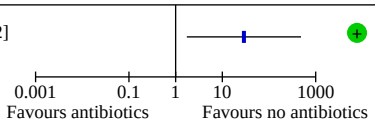
Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Weight	Risk Ratio M-H, Random, 95% CI
Beaton 2022	68	409	0	409	100.0%	137.00 [8.51 , 2205.07]
Brick 1950	0	38	0	38		Not estimable
Total (95% CI)		**447**		**447**	**100.0%**	**137.00 [8.51 , 2205.07]**
Total events:	68		0			

Heterogeneity: Not applicable
Test for overall effect: Z = 3.47 (P = 0.0005)
Test for subgroup differences: Not applicable



Analysis 1.10. Comparison 1: Comparison 1 - Antibiotic prophylaxis versus no antibiotic prophylaxis, Outcome 10: Adverse events: local reactions to injection

Study or Subgroup	Antibiotics Events	Antibiotics Total	No antibiotics Events	No antibiotics Total	Risk Ratio M-H, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G
Beaton 2022	14	409	0	409	29.00 [1.74 , 484.52]	+	+	−	+	+	+	+



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

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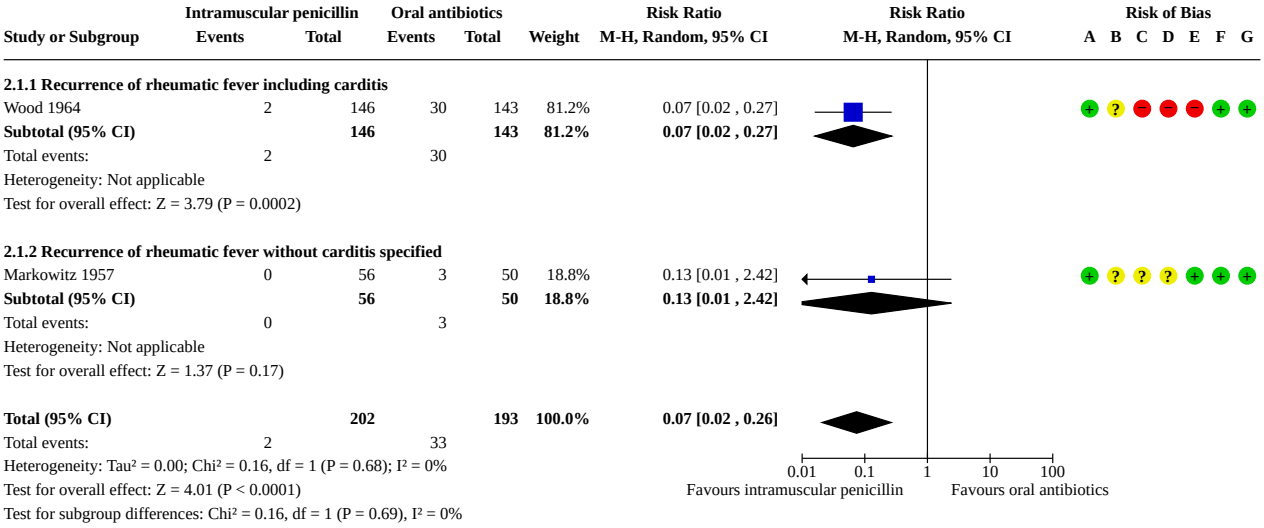
Comparison 2. Comparison 2 - Intramuscular penicillin versus oral antibiotic

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Recurrence of rheumatic fever (count, risk ratio)	2	395	Risk Ratio (M-H, Random, 95% CI)	0.07 [0.02, 0.26]
2.1.1 Recurrence of rheumatic fever including carditis	1	289	Risk Ratio (M-H, Random, 95% CI)	0.07 [0.02, 0.27]
2.1.2 Recurrence of rheumatic fever without carditis specified	1	106	Risk Ratio (M-H, Random, 95% CI)	0.13 [0.01, 2.42]
2.2 Recurrence of rheumatic fever (rate, rate ratio)	2		Rate Ratio (IV, Random, 95% CI)	0.12 [0.04, 0.33]
2.3 Progression of rheumatic heart disease	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
2.4 All-cause mortality	1		Peto Odds Ratio (Peto, Fixed, 95% CI)	Totals not selected
2.5 Poor treatment adherence	2	577	Risk Ratio (M-H, Random, 95% CI)	0.12 [0.06, 0.22]

Analysis 2.1. Comparison 2: Comparison 2 - Intramuscular penicillin versus oral antibiotic, Outcome 1: Recurrence of rheumatic fever (count, risk ratio)



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

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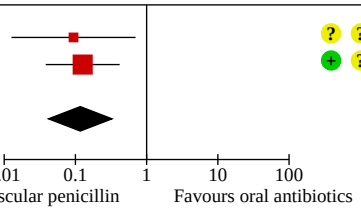
Analysis 2.2. Comparison 2: Comparison 2 - Intramuscular penicillin versus oral antibiotic, Outcome 2: Recurrence of rheumatic fever (rate, rate ratio)

Study or Subgroup	log[Rate Ratio]	SE	Weight	Rate Ratio IV, Random, 95% CI	Risk of Bias A B C D E F G
Feinstein 1959	-2.364155	1.023533	26.2%	0.09 [0.01 , 0.70]	? ? − − + + +
Feinstein 1968	-2.067642	0.60975	73.8%	0.13 [0.04 , 0.42]	+ ? − − − + +
Total (95% CI)			**100.0%**	**0.12 [0.04 , 0.33]**	

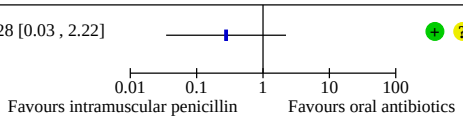
Heterogeneity: Tau² = 0.00; Chi² = 0.06, df = 1 (P = 0.80); I² = 0%
Test for overall effect: Z = 4.10 (P < 0.0001)
Test for subgroup differences: Not applicable



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 2.3. Comparison 2: Comparison 2 - Intramuscular penicillin versus oral antibiotic, Outcome 3: Progression of rheumatic heart disease

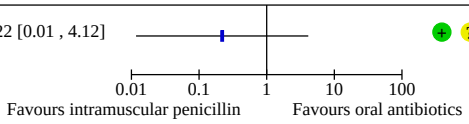
Study or Subgroup	Intramuscular penicillin Events	Intramuscular penicillin Total	Oral antibiotics Events	Oral antibiotics Total	Risk Ratio M-H, Random, 95% CI	Risk of Bias A B C D E F G
Wood 1964	1	87	7	169	0.28 [0.03 , 2.22]	+ ? − − − + +



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 2.4. Comparison 2: Comparison 2 - Intramuscular penicillin versus oral antibiotic, Outcome 4: All-cause mortality

Study or Subgroup	Intramuscular penicillin Events	Intramuscular penicillin Total	Oral antibiotics Events	Oral antibiotics Total	Peto Odds Ratio Peto, Fixed, 95% CI	Risk of Bias A B C D E F G
Wood 1964	0	146	2	285	0.22 [0.01 , 4.12]	+ ? − − − + +



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

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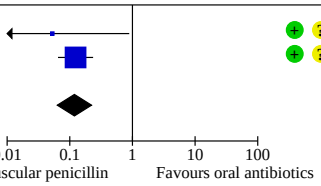
Analysis 2.5. Comparison 2: Comparison 2 - Intramuscular penicillin versus oral antibiotic, Outcome 5: Poor treatment adherence

Study or Subgroup	Intramuscular penicillin Events	Intramuscular penicillin Total	Oral antibiotics Events	Oral antibiotics Total	Weight	Risk Ratio M-H, Random, 95% CI	Risk of Bias A	B	C	D	E	F	G
Markowitz 1957	0	56	8	50	4.9%	0.05 [0.00 , 0.89]	+	?	?	?	+	+	+
Wood 1964	9	156	147	315	95.1%	0.12 [0.06 , 0.24]	+	?	−	−	−	+	+
Total (95% CI)		**212**		**365**	**100.0%**	**0.12 [0.06 , 0.22]**							
Total events:	9		155										

Heterogeneity: Tau² = 0.00; Chi² = 0.33, df = 1 (P = 0.56); I² = 0%
Test for overall effect: Z = 6.65 (P < 0.00001)
Test for subgroup differences: Not applicable



Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

APPENDICES

Appendix 1. Search strategies

CENTRAL

Date run: 10 March 2024

#1 ([mh "Rheumatic Fever"] OR [mh "Rheumatic Nodule"] OR [mh "Wissler's Syndrome"] OR (Acute Articular Rheumatism* OR Acute Rheumati* OR Aschoff Bodies OR Bouillaud* OR Hyperergic Subsepsis OR Inflammatory Rheumati* OR Polyarthritis Rheumatica OR Polyarthritis Rheumatica Acuta OR Polyarthritis Rheumaticas OR Rheumatic Arthritides OR Rheumatic Arthriti* OR Rheumatic Card* OR Rheumatic Fever* OR Rheumatic Heart OR Rheumatic Nodule* OR Rheumatoid Fever* OR Rheumatic Valv* OR Rheumatoid Polyarthritis OR Subsepsis Allergica OR Subsepsis Hyperergica OR Wissler* OR Fereol Node OR Meynet Node OR Rheumatic Skin):ti,ab) AND ([mh "Anti-Bacterial Agents"] OR [mh "Antibiotic Prophylaxis"] OR [mh "Penicillin G Benzathine"] OR [mh "Penicillin V"] OR [mh Sulfadiazine] OR [mh Macrolides] OR [mh Azithromycin] OR (Antibacterial OR Anti-Bacterial OR Antibiotic* OR Antimycobacterial OR Anti-Mycobacterial OR Bacteriocid* OR Penicillin G Benzathine OR Aminopenil OR Banzacillin OR Bayer 5371 OR Beacillin OR Benacil OR Bencelin OR Benzacillin OR Benzanil Simple OR Benzanthine Penicillin OR Benzathin Penicillin OR Benzathine Benzylpenicillin* OR Benzathine Penicillin* OR Benzatine Penicillin OR Benzetacil OR Benzetacil OR Benzethacil OR Benzilfan OR Benzothine Penicillin OR Benzylpenicillin Benzathin OR Bicillin OR Brevicilina OR Cepacilina OR Cillenta OR Debecillin OR Debecyclin* OR Debecylin OR Debycillin OR Diaminocillina OR Dibencil* OR Durabiotic OR Duropenin OR Extencillin* OR Isoject Permapen OR Lentocillin OR Lentopenil OR Liquocillin OR Longacilina OR Longicid OR Lutecilina OR Moldamin OR Neolin OR "Pen Di Ben" OR Penadur OR Pencom OR Pendepon OR "Pendi Ben" OR Penduran OR Penduzan OR Pendysin OR Penicillin Benzathine OR Penicillindamin OR Penidural OR Penidure OR Penilente OR "Peniroger Retard" OR Penretard OR Permapen OR Pheliquin* OR "Provipen Benzatina" OR Tardocillin OR Tripenadur OR "Wycillina A P" OR Zalpen OR "Penicillin V" OR "Abbocillin VK" OR Anapenil OR "Apo Pen VK" OR Apocillin OR "Apopen VK" OR Arcasin OR Beapen OR "Benzathine Phenoxy Methylpenicillin" OR "Benzathine Phenoxymethylpenicillin" OR Beromycin OR Betapen OR "Bicillin 5" OR Calciopen OR "Calcipen V" OR "Cilicaine VK" OR "Compocillin Vk" OR "Compocillin-V" OR "Distaquaine V K" OR Fenocin OR Fenoxymethylpenicillin OR Fenoxypen OR Icipen OR Isocillin OR Kavepenin OR Kavipen OR "Ledercillin VK" OR "Len V.K." OR Megacilin* OR Milcopen OR "Nadopen V" OR Newcillin OR "Novo VK" OR "Novopen VK" OR "Oracillin VK" OR Orapen OR "Ospa V" OR Ostrocilline OR "PC Pen Vk" OR "Pen Vee for Oral Suspension" OR "Pen Vee K" OR "Pen Vee Suspension" OR "Pen Vi K" OR "Pen VK" OR "Penapar VK" OR "Penavlon V" OR Penbeta OR Penicillin Beromycin OR Penicillin Berromycin OR "Penicillin 5" OR "Penicillin VK" OR Pentranex OR "Pen-Vee Oral" OR Penvikal OR "Pfizerpen VK" OR "Phenoxymethyl Penicillin" OR Phenoxymethylpenicillin OR "Rafapen V K" OR "Robicillin VK" OR "Rocilin VK" OR "Servipen V" OR "Trepopen VK" OR "Truxcillin VK" OR "Uticillin VK" OR "V Cil K" OR V Cillin K OR "VCillin K" OR Veetids OR Vegacillin OR Vepicombin OR "V-Kal-K" OR "V-Penicillin Kalium" OR Sulfadiazine OR "2 Sulfanil Amidopyrimidine" OR "2 Sulfanilamidopyrimidine" OR Adiazine OR Aldiazine OR Cocodiazine OR "Coco-Diazine" OR Codiazine OR Cremodiazine OR Debenal OR "Di Azo Mil" OR Diastrep OR Diazine OR Eskadiazine OR Eustral OR Keladiazine OR Liquadiazine OR Microsulfon OR Pirimal OR Pyrimal OR Sterazine OR Sulfacombin OR Sulfadiazin* OR Sulfapyrimidine OR Sulfazin* OR Sulphadiazine OR Macrolide* OR Macrotetrolide* OR Azithromycin OR Azalide* OR Aruzilina OR Atizor OR Azadose OR Azasite OR Azatril OR Azenil OR Azibiot OR Azimin OR Azithral OR Azitrocin OR Azitrocin OR Azitromax OR Azitromicin OR Azitromicina OR Aziwok OR Azomyne OR Aztrin OR Azydrop OR Azyter OR Azythromycin OR Azythromycin OR Bazyt OR "CP 62993" OR CP62993 OR Forcin OR Goxal OR Inedol OR Infectoazit OR "ISV 401" OR ISV401 OR Kromicin OR Macrozit OR Mezatrin OR Octavax OR Ordipha

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OR Ribotrex OR Sumamed OR Sunamed OR Tobyl OR Toraseptol OR Tromix OR Trozocina OR Ultreon OR Vinzam OR Xithrone OR "XZ 450" OR XZ450 OR Zaret OR Zarom OR Zentavion OR Zetamax OR Zeto OR Zibramax OR Zifin OR Zimericina OR Zistic OR Zithromax OR Zithrox OR Zitinn OR Zitrim OR Zitrobifan OR Zitrocin OR Zitromax OR Zmax):ti,ab)

Trials: 113

MEDLINE

Database: Ovid MEDLINE(R) ALL <1946 to 10 March 2024>

1 Rheumatic Fever/ or Rheumatic Nodule/ or Wissler's Syndrome/ or (Acute Articular Rheumatism* or Acute Rheumati* or Aschoff Bodies or Bouillaud* or Hyperergic Subsepsis or Inflammatory Rheumati* or Polyarthritis Rheumatica or Polyarthritis Rheumatica Acuta or Polyarthritis Rheumaticas or Rheumatic Arthritides or Rheumatic Arthriti* or Rheumatic Card* or Rheumatic Fever* or Rheumatic Heart or Rheumatic Nodule* or Rheumatoid Fever* or Rheumatic Valv* or Rheumatoid Polyarthritis or Subsepsis Allergica or Subsepsis Hyperergica or Wissler* or Fereol Node or Meynet Node or Rheumatic Skin).ti,ab. (23703)

2 Anti-Bacterial Agents/ or Antibiotic Prophylaxis/ or Penicillin G Benzathine/ or Penicillin V/ or Sulfadiazine/ or Macrolides/ or Azithromycin/ or (Antibacterial or Anti-Bacterial or Antibiotic* or Antimycobacterial or Anti-Mycobacterial or Bacteriocid* or Penicillin G Benzathine or Aminopenil or Banzacillin or Bayer 5371 or Beacillin or Benacil or Bencelin or Benzacillin or Benzanil Simple or Benzanthine Penicillin or Benzathin Penicillin or Benzathine Benzylpenicillin* or Benzathine Penicillin* or Benzatine Penicillin or Benzetacil or Benzetacil or Benzethacil or Benzilfan or Benzothine Penicillin or Benzylpenicillin Benzathin or Bicillin or Brevicilina or Cepacilina or Cillenta or Debecillin or Debecyclin* or Debecylin or Debycillin or Diaminocillina or Dibencil* or Durabiotic or Duropenin or Extencillin* or Isoject Permapen or Lentocillin or Lentopenil or Liquocillin or Longacilina or Longicid or Lutecilina or Moldamin or Neolin or "Pen Di Ben" or Penadur or Pencom or Pendepon or "Pendi Ben" or Penduran or Penduzan or Pendysin or Penicillin Benzathine or Penicillindamin or Penidural or Penidure or Penilente or "Peniroger Retard" or Penretard or Permapen or Pheliquin* or "Provipen Benzatina" or Tardocillin or Tripenadur or "Wycillina A P" or Zalpen or "Penicillin V" or "Abbocillin VK" or Anapenil or "Apo Pen VK" or Apocillin or "Apopen VK" or Arcasin or Beapen or "Benzathine Phenoxy Methylpenicillin" or "Benzathine Phenoxymethylpenicillin" or Beromycin or Betapen or "Bicillin 5" or Calciopen or "Calcipen V" or "Cilicaine VK" or "Compocillin Vk" or "Compocillin-V" or "Distaquaine V K" or Fenocin or Fenoxymethylpenicillin or Fenoxypen or Icipen or Isocillin or Kavepenin or Kavipen or "Ledercillin VK" or "Len V.K." or Megacilin* or Milcopen or "Nadopen V" or Newcillin or "Novo VK" or "Novopen VK" or "Oracillin VK" or Orapen or "Ospa V" or Ostrocilline or "PC Pen Vk" or "Pen Vee for Oral Suspension" or "Pen Vee K" or "Pen Vee Suspension" or "Pen Vi K" or "Pen VK" or "Penapar VK" or "Penavlon V" or Penbeta or Penicillin Beromycin or Penicillin Berromycin or "Penicillin 5" or "Penicillin VK" or Pentranex or "Pen-Vee Oral" or Penvikal or "Pfizerpen VK" or "Phenoxymethyl Penicillin" or Phenoxymethylpenicillin or "Rafapen V K" or "Robicillin VK" or "Rocilin VK" or "Servipen V" or "Trepopen VK" or "Truxcillin VK" or "Uticillin VK" or "V Cil K" or V Cillin K or "VCillin K" or Veetids or Vegacillin or Vepicombin or "V-Kal-K" or "V-Penicillin Kalium" or Sulfadiazine or "2 Sulfanil Amidopyrimidine" or "2 Sulfanilamidopyrimidine" or Adiazine or Aldiazine or Cocodiazine or "Coco-Diazine" or Codiazine or Cremodiazine or Debenal or "Di Azo Mil" or Diastrep or Diazine or Eskadiazine or Eustral or Keladiazine or Liquadiazine or Microsulfon or Pirimal or Pyrimal or Sterazine or Sulfacombin or Sulfadiazin* or Sulfapyrimidine or Sulfazin* or Sulphadiazine or Macrolide* or Macrotetrolide* or Azithromycin or Azalide* or Aruzilina or Atizor or Azadose or Azasite or Azatril or Azenil or Azibiot or Azimin or Azithral or Azitrocin or Azitrocin or Azitromax or Azitromicin or Azitromicina or Aziwok or Azomyne or Aztrin or Azydrop or Azyter or Azythromycin or Azythromycin or Bazyt or "CP 62993" or CP62993 or Forcin or Goxal or Inedol or Infectoazit or "ISV 401" or ISV401 or Kromicin or Macrozit or Mezatrin or Octavax or Ordipha or Ribotrex or Sumamed or Sunamed or Tobyl or Toraseptol or Tromix or Trozocina or Ultreon or Vinzam or Xithrone or "XZ 450" or XZ450 or Zaret or Zarom or Zentavion or Zetamax or Zeto or Zibramax or Zifin or Zimericina or Zistic or Zithromax or Zithrox or Zitinn or Zitrim or Zitrobifan or Zitrocin or Zitromax or Zmax).ti,ab. (730562)

3 1 and 2 (1538)

4 exp Animals/ not Humans.sh. (5201238)

5 3 not 4 (1534)

Embase

Database: Embase <1974 to 10 March 2024>

1 *Rheumatic Fever/ or *Rheumatic Heart Disease/ or *Rheumatoid Nodule/ or (Acute Articular Rheumatism* or Acute Rheumati* or Aschoff Bodies or Bouillaud* or Hyperergic Subsepsis or Inflammatory Rheumati* or Polyarthritis Rheumatica or Polyarthritis Rheumatica Acuta or Polyarthritis Rheumaticas or Rheumatic Arthritides or Rheumatic Arthriti* or Rheumatic Card* or Rheumatic Fever* or Rheumatic Heart or Rheumatic Nodule* or Rheumatoid Fever* or Rheumatic Valv* or Rheumatoid Polyarthritis or Subsepsis Allergica or Subsepsis Hyperergica or Wissler* or Fereol Node or Meynet Node or Rheumatic Skin).ti,ab. (23660)

2 *Antibiotic Agent/ or *Antibiotic Prophylaxis/ or *Benzathine Penicillin/ or *Benzathine Penicillin V/ or *Penicillin V/ or *Penicillin V Calcium/ or *Penicillin V Potassium/ or *Sulfadiazine/ or *Macrolide/ or *Azithromycin/ or (Antibacterial or Anti-Bacterial or Antibiotic* or Antimycobacterial or Anti-Mycobacterial or Bacteriocid* or Penicillin G Benzathine or Aminopenil or Banzacillin or Bayer 5371 or Beacillin or Benacil or Bencelin or Benzacillin or Benzanil Simple or Benzanthine Penicillin or Benzathin Penicillin or Benzathine Benzylpenicillin* or Benzathine Penicillin* or Benzatine Penicillin or Benzetacil or Benzetacil or Benzethacil or Benzilfan or Benzothine Penicillin or

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Benzylpenicillin Benzathin or Bicillin or Brevicilina or Cepacilina or Cillenta or Debecillin or Debecyclin* or Debecylin or Debycillin or Diaminocillina or Dibencil* or Durabiotic or Duropenin or Extencillin* or Isoject Permapen or Lentocillin or Lentopenil or Liquocillin or Longacilina or Longicid or Lutecilina or Moldamin or Neolin or "Pen Di Ben" or Penadur or Pencom or Pendepon or "Pendi Ben" or Penduran or Penduzan or Pendysin or Penicillin Benzathine or Penicillindamin or Penidural or Penidure or Penilente or "Peniroger Retard" or Penretard or Permapen or Pheliquin* or "Provipen Benzatina" or Tardocillin or Tripenadur or "Wycillina A P" or Zalpen or "Penicillin V" or "Abbocillin VK" or Anapenil or "Apo Pen VK" or Apocillin or "Apopen VK" or Arcasin or Beapen or "Benzathine Phenoxy Methylpenicillin" or "Benzathine Phenoxymethylpenicillin" or Beromycin or Betapen or "Bicillin 5" or Calciopen or "Calcipen V" or "Cilicaine VK" or "Compocillin Vk" or "Compocillin-V" or "Distaquaine V K" or Fenocin or Fenoxymethylpenicillin or Fenoxypen or Icipen or Isocillin or Kavepenin or Kavipen or "Ledercillin VK" or "Len V.K." or Megacilin* or Milcopen or "Nadopen V" or Newcillin or "Novo VK" or "Novopen VK" or "Oracillin VK" or Orapen or "Ospa V" or Ostrocilline or "PC Pen Vk" or "Pen Vee for Oral Suspension" or "Pen Vee K" or "Pen Vee Suspension" or "Pen Vi K" or "Pen VK" or "Penapar VK" or "Penavlon V" or Penbeta or Penicillin Beromycin or Penicillin Berromycin or "Penicillin 5" or "Penicillin VK" or Pentranex or "Pen-Vee Oral" or Penvikal or "Pfizerpen VK" or "Phenoxymethyl Penicillin" or Phenoxymethylpenicillin or "Rafapen V K" or "Robicillin VK" or "Rocilin VK" or "Servipen V" or "Trepopen VK" or "Truxcillin VK" or "Uticillin VK" or "V Cil K" or V Cillin K or "VCillin K" or Veetids or Vegacillin or Vepicombin or "V-Kal-K" or "V-Penicillin Kalium" or Sulfadiazine or "2 Sulfanil Amidopyrimidine" or "2 Sulfanilamidopyrimidine" or Adiazine or Aldiazine or Cocodiazine or "Coco-Diazine" or Codiazine or Cremodiazine or Debenal or "Di Azo Mil" or Diastrep or Diazine or Eskadiazine or Eustral or Keladiazine or Liquadiazine or Microsulfon or Pirimal or Pyrimal or Sterazine or Sulfacombin or Sulfadiazin* or Sulfapyrimidine or Sulfazin* or Sulphadiazine or Macrolide* or Macrotetrolide* or Azithromycin or Azalide* or Aruzilina or Atizor or Azadose or Azasite or Azatril or Azenil or Azibiot or Azimin or Azithral or Azitrocin or Azitrocin or Azitromax or Azitromicin or Azitromicina or Aziwok or Azomyne or Aztrin or Azydrop or Azyter or Azythromycin or Azythromycin or Bazyt or "CP 62993" or CP62993 or Forcin or Goxal or Inedol or Infectoazit or "ISV 401" or ISV401 or Kromicin or Macrozit or Mezatrin or Octavax or Ordipha or Ribotrex or Sumamed or Sunamed or Tobyl or Toraseptol or Tromix or Trozocina or Ultreon or Vinzam or Xithrone or "XZ 450" or XZ450 or Zaret or Zarom or Zentavion or Zetamax or Zeto or Zibramax or Zifin or Zimericina or Zistic or Zithromax or Zithrox or Zitinn or Zitrim or Zitrobifan or Zitrocin or Zitromax or Zmax).ti,ab. (744241)

3 1 and 2 (1688)

4 (rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset$1).ti. and animal experiment/ (1243179)

5 Animal experiment/ not (human experiment/ or human/) (2612732)

6 4 or 5 (2684773)

7 3 not 6 (1682)

8 limit 7 to medline (198)

9 7 not 8 (1484)

CPCI-S

11 March 2024

1: Acute Articular Rheumatism* OR Acute Rheumati* OR Aschoff Bodies OR Bouillaud* OR Hyperergic Subsepsis OR Inflammatory Rheumati* OR Polyarthritis Rheumatica OR Polyarthritis Rheumatica Acuta OR Polyarthritis Rheumaticas OR Rheumatic Arthritides OR Rheumatic Arthriti* OR Rheumatic Card* OR Rheumatic Fever* OR Rheumatic Heart OR Rheumatic Nodule* OR Rheumatoid Fever* OR Rheumatic Valv* OR Rheumatoid Polyarthritis OR Subsepsis Allergica OR Subsepsis Hyperergica OR Wissler* OR Fereol Node OR Meynet Node OR Rheumatic Skin (Title) OR Acute Articular Rheumatism* OR Acute Rheumati* OR Aschoff Bodies OR Bouillaud* OR Hyperergic Subsepsis OR Inflammatory Rheumati* OR Polyarthritis Rheumatica OR Polyarthritis Rheumatica Acuta OR Polyarthritis Rheumaticas OR Rheumatic Arthritides OR Rheumatic Arthriti* OR Rheumatic Card* OR Rheumatic Fever* OR Rheumatic Heart OR Rheumatic Nodule* OR Rheumatoid Fever* OR Rheumatic Valv* OR Rheumatoid Polyarthritis OR Subsepsis Allergica OR Subsepsis Hyperergica OR Wissler* OR Fereol Node OR Meynet Node OR Rheumatic Skin (Abstract) Editions: WOS.ISTP Date Run: Mon Mar 11 2024 09:20:04 GMT+0000 (Greenwich Mean Time) Results: 29583
2: Antibacterial OR Anti-Bacterial OR Antibiotic* OR Antimycobacterial OR Anti-Mycobacterial OR Bacteriocid* OR Penicillin G Benzathine OR Aminopenil OR Banzacillin OR Bayer 5371 OR Beacillin OR Benacil OR Bencelin OR Benzacillin OR Benzanil Simple OR Benzanthine Penicillin OR Benzathin Penicillin OR Benzathine Benzylpenicillin* OR Benzathine Penicillin* OR Benzatine Penicillin OR Benzetacil OR Benzetacil OR Benzethacil OR Benzilfan OR Benzothine Penicillin OR Benzylpenicillin Benzathin OR Bicillin OR Brevicilina OR Cepacilina OR Cillenta OR Debecillin OR Debecyclin* OR Debecylin OR Debycillin OR Diaminocillina OR Dibencil* OR Durabiotic OR Duropenin OR Extencillin* OR Isoject Permapen OR Lentocillin OR Lentopenil OR Liquocillin OR Longacilina OR Longicid OR Lutecilina OR Moldamin OR Neolin OR "Pen Di Ben" OR Penadur OR Pencom OR Pendepon OR "Pendi Ben" OR Penduran OR Penduzan OR Pendysin OR Penicillin Benzathine OR Penicillindamin OR Penidural OR Penidure OR Penilente OR "Peniroger Retard" OR Penretard OR Permapen OR Pheliquin* OR "Provipen Benzatina" OR Tardocillin OR Tripenadur OR "Wycillina A P" OR Zalpen OR "Penicillin V" OR "Abbocillin VK" OR Anapenil OR "Apo Pen VK" OR Apocillin OR "Apopen VK" OR Arcasin OR Beapen OR "Benzathine Phenoxy Methylpenicillin" OR "Benzathine Phenoxymethylpenicillin" OR Beromycin OR Betapen OR "Bicillin 5" OR Calciopen OR "Calcipen V" OR "Cilicaine VK" OR "Compocillin Vk"

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OR "Compocillin-V" OR "Distaquaine V K" OR Fenocin OR Fenoxymethylpenicillin OR Fenoxypen OR Icipen OR Isocillin OR Kavepenin OR Kavipen OR "Ledercillin VK" OR "Len V.K." OR Megacilin* OR Milcopen OR "Nadopen V" OR Newcillin OR "Novo VK" OR "Novopen VK" OR "Oracillin VK" OR Orapen OR "Ospa V" OR Ostrocilline OR "PC Pen Vk" OR "Pen Vee for Oral Suspension" OR "Pen Vee K" OR "Pen Vee Suspension" OR "Pen Vi K" OR "Pen VK" OR "Penapar VK" OR "Penavlon V" OR Penbeta OR Penicillin Beromycin OR Penicillin Berromycin OR "Penicillin 5" OR "Penicillin VK" OR Pentranex OR "Pen-Vee Oral" OR Penvikal OR "Pfizerpen VK" OR "Phenoxymethyl Penicillin" OR Phenoxymethylpenicillin OR "Rafapen V K" OR "Robicillin VK" OR "Rocilin VK" OR "Servipen V" OR "Trepopen VK" OR "Truxcillin VK" OR "Uticillin VK" OR "V Cil K" OR V Cillin K OR "VCillin K" OR Veetids OR Vegacillin OR Vepicombin OR "V-Kal-K" OR "V-Penicillin Kalium" OR Sulfadiazine OR "2 Sulfanil Amidopyrimidine" OR "2 Sulfanilamidopyrimidine" OR Adiazine OR Aldiazine OR Cocodiazine OR "Coco-Diazine" OR Codiazine OR Cremodiazine OR Debenal OR "Di Azo Mil" OR Diastrep OR Diazine OR Eskadiazine OR Eustral OR Keladiazine OR Liquadiazine OR Microsulfon OR Pirimal OR Pyrimal OR Sterazine OR Sulfacombin OR Sulfadiazin* OR Sulfapyrimidine OR Sulfazin* OR Sulphadiazine OR Macrolide* OR Macrotetrolide* OR Azithromycin OR Azalide* OR Aruzilina OR Atizor OR Azadose OR Azasite OR Azatril OR Azenil OR Azibiot OR Azimin OR Azithral OR Azitrocin OR Azitrocin OR Azitromax OR Azitromicin OR Azitromicina OR Aziwok OR Azomyne OR Aztrin OR Azydrop OR Azyter OR Azythromycin OR Azythromycin OR Bazyt OR "CP 62993" OR CP62993 OR Forcin OR Goxal OR Inedol OR Infectoazit OR "ISV 401" OR ISV401 OR Kromicin OR Macrozit OR Mezatrin OR Octavax OR Ordipha OR Ribotrex OR Sumamed OR Sunamed OR Tobyl OR Toraseptol OR Tromix OR Trozocina OR Ultreon OR Vinzam OR Xithrone OR "XZ 450" OR XZ450 OR Zaret OR Zarom OR Zentavion OR Zetamax OR Zeto OR Zibramax OR Zifin OR Zimericina OR Zistic OR Zithromax OR Zithrox OR Zitinn OR Zitrim OR Zitrobifan OR Zitrocin OR Zitromax OR Zmax (Title) OR Antibacterial OR Anti-Bacterial OR Antibiotic* OR Antimycobacterial OR Anti-Mycobacterial OR Bacteriocid* OR Penicillin G Benzathine OR Aminopenil OR Banzacillin OR Bayer 5371 OR Beacillin OR Benacil OR Bencelin OR Benzacillin OR Benzanil Simple OR Benzanthine Penicillin OR Benzathin Penicillin OR Benzathine Benzylpenicillin* OR Benzathine Penicillin* OR Benzatine Penicillin OR Benzetacil OR Benzetacil OR Benzethacil OR Benzilfan OR Benzothine Penicillin OR Benzylpenicillin Benzathin OR Bicillin OR Brevicilina OR Cepacilina OR Cillenta OR Debecillin OR Debecyclin* OR Debecylin OR Debycillin OR Diaminocillina OR Dibencil* OR Durabiotic OR Duropenin OR Extencillin* OR Isoject Permapen OR Lentocillin OR Lentopenil OR Liquocillin OR Longacilina OR Longicid OR Lutecilina OR Moldamin OR Neolin OR "Pen Di Ben" OR Penadur OR Pencom OR Pendepon OR "Pendi Ben" OR Penduran OR Penduzan OR Pendysin OR Penicillin Benzathine OR Penicillindamin OR Penidural OR Penidure OR Penilente OR "Peniroger Retard" OR Penretard OR Permapen OR Pheliquin* OR "Provipen Benzatina" OR Tardocillin OR Tripenadur OR "Wycillina A P" OR Zalpen OR "Penicillin V" OR "Abbocillin VK" OR Anapenil OR "Apo Pen VK" OR Apocillin OR "Apopen VK" OR Arcasin OR Beapen OR "Benzathine Phenoxy Methylpenicillin" OR "Benzathine Phenoxymethylpenicillin" OR Beromycin OR Betapen OR "Bicillin 5" OR Calciopen OR "Calcipen V" OR "Cilicaine VK" OR "Compocillin Vk" OR "Compocillin-V" OR "Distaquaine V K" OR Fenocin OR Fenoxymethylpenicillin OR Fenoxypen OR Icipen OR Isocillin OR Kavepenin OR Kavipen OR "Ledercillin VK" OR "Len V.K." 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3: #2 AND #1

Results: 41 records

ClinicalTrials.gov

10 March 2024

Advanced Search

Condition or disease: Rheumatic Fever OR Rheumatic Nodule OR Wissler's Syndrome OR Inflammatory Rheumatism OR Rheumatic Arthritis OR Rheumatic Cardiac OR Rheumatic Heart

Intervention/treatment: Antibacterial OR Antibiotics OR Penicillin G Benzathine OR Penicillin V OR Sulfadiazine OR Macrolide OR Azithromycin

55 studies found

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WHO

10 March 2024

Advanced Search

Rheumatic Fever OR Rheumatic Nodule OR Wissler's Syndrome OR Inflammatory Rheumatism OR Rheumatic Arthritis OR Rheumatic Cardiac OR Rheumatic Heart in the Condition

Anti-Bacterial OR Antibacterial OR Antibiotic OR Antibiotics OR Penicillin G Benzathine OR Penicillin V OR Sulfadiazine OR Macrolides OR Macrolide OR Azithromycin in the Intervention

Recruitment status is ALL

6 records for 6 trials found

ISRCTN.com

10 March 2024

3 records found

CONTRIBUTIONS OF AUTHORS

JJHB designed and co-ordinated the review, developed and wrote the original protocol, assessed studies for inclusion and resolved disagreements, extracted data, assessed risk of bias and resolved disagreements, analysed the data, graded the certainty of the evidence, interpreted the data, wrote the Abstract, Plain language summary, Results, Conclusion and part of the Background, Methods and Discussion sections, edited the review in response to feedback, addressed peer-review queries and was responsible for project administration.

ST assessed studies for inclusion, extracted data, assessed risk of bias, helped write part of the Results and Discussion, and edited and provided advice on the manuscript.

SS helped write part of the Discussion, and edited and provided advice on the manuscript.

SAA assessed studies for inclusion, extracted data, wrote part of the information regarding included studies, and edited and provided advice on the manuscript.

JY assessed studies for inclusion, extracted data, helped format references for excluded studies, and edited and provided advice on the manuscript.

MS assessed studies for inclusion, and edited and provided advice on the manuscript.

MA assisted in the co-ordination of tasks, developed and wrote the original protocol, assessed studies for inclusion, resolved disagreements regarding study eligibility and extracted data, assessed risk of bias and resolved disagreements, graded the certainty of the evidence, interpreted the data, and edited and provided advice on the manuscript.

FP helped write part of the Background and Methods, and edited and provided advice on the manuscript.

HG assessed risk of bias, and edited and provided advice on the manuscript.

FS performed the literature search, and edited and provided advice on the manuscript.

NA edited and provided advice on the manuscript.

MC edited and provided advice on the manuscript.

EM edited and provided advice on the manuscript.

DC edited and provided advice on the manuscript.

RP acted as the main senior author (providing advice on a weekly basis), interpreted the data, arbitrated disagreements in the grading of the certainty of the evidence, and edited and provided advice on the manuscript.

DECLARATIONS OF INTEREST

JJHB works as an Internal Medicine Trainee at Oxford University Hospitals Trust. He has no conflicts of interest.

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ST works as an Internal Medicine Trainee at University Hospitals Birmingham NHS Foundation Trust. She has no conflicts of interest.

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MA works as a Cardiology Consultant at Barts Heart Centre. He has no conflicts of interest.

FP works as a health professional at UCL, University of London. He has no conflicts of interest.

HG works as an Internal Medicine Trainee at The University Hospital Monklands. He has no conflicts of interest.

FS is the Director at Systematic Review Consultants LTD, Evidence Synthesis Manager at the University of Oxford and Senior Research Associate at the University of Bristol. He holds an honorary assistant professorship from the University of Nottingham and an honorary lectureship from University College London. He was an Information Specialist for Cochrane Schizophrenia, Cochrane Neuromuscular, Cochrane Gut, Cochrane Heart and Cochrane Developmental, Psychosocial and Learning Problems. FS was not involved in the editorial process for the review. He has no conflicts of interest.

NA works as a Cardiology Registrar at Barts Heart Centre. She has no conflicts of interest.

MC is a Consultant Cardiologist at São Tomé and Príncipe Hospital. She has no conflicts of interest.

EM is a Professor of Cardiology at Paris City University and works as a health professional for APHP and INSERM. He declares grants from Abbott Fund, Boston Scientific Corporation, Medtronic USA Inc and Zoll Medical Corporation (personal payments). He has published opinions on the topic for INSERM.

DC is the Head of the Discipline of Cardiology and Cardiology Consultant at The University of Sydney. He reports conducting studies eligible for inclusion in the review; these are academic papers only, none of which had any external funding. DC was not involved in decisions regarding the eligibility of these studies, extracting data, assessing risk of bias or grading the certainty of the evidence. These tasks were performed by at least two independent review authors (JJHB, SAA, MS, ST, MA, JY). He has no conflicts of interest.

RP is a Professor of Cardiology at University College of London and works as a Cardiology Consultant - Cardiac Electrophysiologist at St Bartholomew's Hospital London, UK. He is the former Co-ordinating Editor of Cochrane Heart and was not involved in the editorial process for this review. RP declares a grant from the World Health Organization for writing eight systematic reviews for the new Rheumatic Heart Disease and Acute Rheumatic Fever guideline, paid to UCL; however, as Principal Investigator for the project, with responsibility for budget decisions, RP did have access to the funds.

SOURCES OF SUPPORT

Internal sources

- None, Other

  N/a

External sources

- WHO guideline support, Other

  Support for conception, design and focus of the review. Support with salary for information specialist. All other authors worked on a voluntary basis.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Rapid review versus standard intervention review

This review was initially intended to be a rapid review; however, following discussions with Cochrane, it was later felt to be more accurately described as a standard intervention review.

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Types of studies

Conference abstracts were included and screened. We have removed "conference abstracts were excluded if not enough data for extraction". We decided to make this change to be compliant with the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards (MECIR 2023).

For some studies, it was not clear whether they were randomised or not. In these cases, we applied a Chi^2 test to estimate the probability that group allocations and baseline characteristics could have occurred by chance to assist us in making our assessment. We did this to clarify the design of studies where this was unclear.

As we believed that there was enough RCT evidence available on this topic, and the methodology of existing non-RCTs is likely to be of lower quality and to introduce potential biases into the review, we opted to include only randomised controlled trials or quasi-randomised trials (Reeves 2023). We conducted a sensitivity analysis of quasi-randomised trials to assess the effect of this.

We excluded theses, book chapters, animal studies and laboratory studies not carried out in a clinical setting. This had been planned in our initial protocol, agreed with the World Health Organization (WHO), but these details did not make it into our published protocol due to an error.

Types of participants

We have included studies in which more than 80% of participants had a diagnosis of rheumatic heart disease (RHD), which is an established proportion to represent a majority (McKenzie 2023).

Types of interventions

We have added the following comparison, which was not in the protocol (Pelone 2022): 'IM penicillin versus oral antibiotics'. We felt that this was a clinically important comparison, and it was requested by the World Health Organization.

In clarification of our protocol (Pelone 2022), we considered medications other than antibiotic prophylaxis eligible as concomitant medications, provided they were applied to all treatment arms.

Search methods

We also used the following additional search methods to ensure a more extensive search.

1. We checked the reference lists of the included studies and other relevant systematic reviews identified for additional references to trials.
2. We examined any relevant retraction statements and errata for included studies.

Types of outcome measures

Recurrence of rheumatic fever was preferentially used when diagnosed using the Jones criteria, in order to be most clinically relevant to modern medicine, but where this was not possible it was defined as per trial protocol, as stated in our protocol.

Selection of studies

We clarified our approach to screening, as described below, as this was unclear in the protocol.

"Dual screening will be performed on all records identified by the search; 90% agreement is required" was changed to "Six review authors (JJHB, SAA, MS, ST, MA, JY) independently classified screened abstracts as eligible or not eligible against our inclusion criteria. Each title was screened by at least two review authors working independently using Rayyan software (Ouzzani 2016). The second screen was performed by JJHB and MA. After retrieval of full-text papers, six review authors (JJHB, SAA, MS, ST, MA, JY) independently screened the full texts and identified trials for inclusion and exclusion, with each full-text paper reviewed by at least two review authors working independently."

Data extraction and management

We collected information on the comparator (form, dosage, type and length of comparator (placebo, standard medical therapy or oral antibiotics)), as well as details on funding sources of studies and any notable conflicts of interests, in order to better characterise the available data.

Assessment of risk of bias in included studies

For risk of bias assessment, we have used RoB 1 rather than RoB 2 as this tool was felt by the authors to be easiest to use and understand, and either tool is allowed by Cochrane.

Measures of treatment effects

As data were available, we used risk ratios as opposed to odds ratios to present our analyses (Sackett 1996; Sinclair 1994), as Section 6.4.1.2 of the *Cochrane Handbook for Systematic Reviews of Interventions* states that, "odds ratios, like odds, are more difficult to interpret"

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and recommends using risk ratio (Higgins 2022). Exceptions to this were outcomes with very low event rates (i.e. < 1%). For such cases, Section 16.9.5 of the *Cochrane Handbook* states that, "at event rates below 1% the Peto one-step odds ratio method [is] the least biased and most powerful method, and provides the best confidence interval coverage" (Higgins 2011b). Consequently, for these outcomes, we have presented our results using this methodology. We did this to ensure compliance with Cochrane methods.

We did not encounter time-to-event outcomes (such as time to recurrence of rheumatic fever). In future updates, we will express these as hazard ratios with 95% CIs, to take account of the censored nature of the data.

We planned to describe any skewed data reported as medians and interquartile ranges, but this was not necessary.

In the protocol, we said we would measure continuous outcomes using MD and SMD, but we did not find any data in this form, so this was not used.

Unit of analysis issues

Cluster-RCTs

We did not encounter any cluster-RCTs but in future updates we would include this study design. To avoid a unit of analysis error, cluster-randomised trials would be analysed at the same level as the allocation, using a summary measurement from each cluster (Higgins 2023b).

Multi-arm trials

For multiple arms with relevant antibiotics of different doses, we planned to combine these, as necessary, in accordance with the relevant comparison.

We have added the following statement to the methods to clarify how we managed included studies with multiple arms and different medications: "When we identified multi-arm trials (e.g. two arms with different antibiotics and a control arm), we combined the antibiotic arms for comparison 1 (antibiotics versus control) and separated the arms for comparison 2 (IM antibiotics versus oral antibiotics) and subgroup analysis, as required."

Multiple follow-up

For trials that reported on more than one follow-up, we analysed outcomes at the longest possible time of follow-up, to utilise the maximum data available.

Assessment of reporting biases

We planned to create and examine a funnel plot to explore possible small-study effects and perform a formal statistical test (Egger's test) for asymmetry of the primary outcomes; however, we were not able to pool 10 or more studies in a meta-analysis. In future updates of this review, these methods will be used.

Data synthesis

As we were able to perform a meta-analysis, we did not need to use the 'nine-point checklist in the new synthesis without meta-analysis (SWIM) guidance', as stated in our protocol (Campbell 2020).

Subgroup analysis and investigation of heterogeneity

The following subgroup analyses were reported in our protocol as being desirable, but were not conducted due to insufficient available information within the literature:

- rheumatic fever with no carditis;
- rheumatic fever with carditis but no RHD;
- rheumatic fever with residual valvular RHD;
- pregnancy;
- moderate or severe valvular RHD; and
- mild/latent valvular RHD.

However, it should be noted that we included latent and moderate/severe RHD as separate outcomes in the review, as we considered latent RHD and severe RHD to be too clinically different to combine in meta-analysis.

Two further subgroup analyses of 'antibiotic class' had been pre-planned and requested by the WHO, but had not made it into the published protocol due to an error, and have been included within the final manuscript.

We sought to conduct a subgroup analysis by duration of treatment. However, due to the continuous nature of the duration of treatment, this was not possible with binary groups; therefore, we used meta-regression, available in Stata 17.0 (see Methods for further information). This further analysis was necessary to interpret the data in this review.

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Sensitivity analysis

It was noted that three studies used oral benzyl penicillin (Cope 1960; Feinstein 1966; Markowitz 1957). As this formulation of penicillin is now rarely used orally, we decided to apply a further sensitivity analysis to investigate the exclusion of these studies.

We also conducted sensitivity analyses for:

- low risk of bias studies (risk of bias varied between studies);
- peer-reviewed publications (some papers were published in non-peer-reviewed publications);
- missing data (one study had missing data);
- diagnosis of rheumatic fever (historical variation in definition); and
- contemporary trials (at peer review it was raised that the WHO guideline aims to inform contemporary practice, therefore we performed sensitivity analysis to only include studies conducted after the year 2000).

Due to points raised during peer review, we performed further sensitivity analyses for:

1. per protocol analysis; and
2. exclusion of quasi-randomised trials.

These additional sensitivity analyses were done to further clarify the data in this review and are consistent with the *Cochrane Handbook for Systematic Reviews of Interventions*.

Summary of findings and assessment of the certainty of the evidence

We have included a summary of findings section within the Methods in which we state the comparisons, outcomes, software used and time points reported in the summary of findings tables, as this is recommended in the *Cochrane Handbook* (Schünemann 2023).

INDEX TERMS

Medical Subject Headings (MeSH)

Administration, Oral; Anti-Bacterial Agents [adverse effects] [therapeutic use]; *Antibiotic Prophylaxis [methods]; *Disease Progression; Injections, Intramuscular; Penicillins [adverse effects] [therapeutic use]; Randomized Controlled Trials as Topic; Recurrence; *Rheumatic Fever [complications] [drug therapy] [prevention & control]; *Rheumatic Heart Disease [prevention & control]; *Secondary Prevention [methods]

MeSH check words

Adolescent; Child; Humans

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